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The Relationship Between Skill Learning and Repetition Priming: Experimental and Computational Analyses

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Skill learning and repetition priming are considered by some to be supported by separate memory systems. The authors examined the relationship between skill learning and priming in 3 experiments using a digit entering task, in which participants were presented with unique and repeated 5-digit strings with controlled sequential structure. Both skill learning and priming were observed across a wide range of skill levels. Performance reflected the effects of learning at 3 different levels of stimulus structure, calling into question a binary dichotomy between item-specific priming and general skill learning. Two computational models were developed which demonstrated that previous dissociations between skill learning and priming can occur within a single memory system. The experimental and computational results are interpreted as suggesting that skill learning and priming should be viewed as 2 aspects of a single incremental learning mechanism.

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The study of memory has been dramatically influenced by the discovery of dissociations in performance elicited by different types of memory tests that tap different kinds or aspects of memory. The most dramatic dissociations occur in patients with amnesia. Such patients are impaired on direct tests of memory such as recall and recognition, which require reference to and explicit remembering of a previous episode, but they often perform normally on indirect tests of memory, which do not require reference to previous study episodes (Gabrieli, 1998). Parallel dissociations have been observed in neurologically intact populations, in which these types of memory performances can be dissociated both functionally and stochastically (Roediger & McDermott, 1993). Within the class of memory performances that are spared in amnesia, two types of memory phenomena have been most thoroughly characterized: skill learning and repetition priming (e.g., Cohen, 1984; Cohen & Eichenbaum, 1993).

Skill learning refers to the gradual improvement of performance with practice that generalizes to a range of stimuli within a domain of processing. The improvement takes the form of increased accuracy, decreased response times over trials, or both. Robust skill learning has been observed in amnesia in various domains, including motor skills (e.g., Brooks & Baddeley, 1976; Corkin, 1968), perceptual skills (e.g., Cohen & Squire, 1980) and cognitive skills (e.g., Cohen, Eichenbaum, Deacedo, & Corkin, 1985; Knowlton, Ramus, & Squire, 1992; Knowlton, Squire, & Gluck, 1994). For example, practice in reading mirror-reversed text leads participants to show increased reading speed for new examples of mirror-reversed text. In amnesia, skill learning occurs despite an inability to explicitly remember previous learning experiences or their specific contents, and despite a lack of insight into the nature of the acquired skill (see Cohen, 1984; Cohen & Eichenbaum, 1993). In normal participants, too, skill learning can sometimes occur independently of explicit remembering of the learning materials (e.g., Kolers, 1976).

Repetition priming (or simply priming) is a facilitation or biasing of performance that occurs for the specific stimuli encountered in a task. Priming is measured as the improvement in performance, in terms of increased accuracy or decreased response times, for previously presented or repeated stimuli as compared to stimuli presented for the first time. Priming often occurs normally in amnesic patients, both for stimuli with preexisting representations such as words (e.g., Graf, Squire, & Mandler, 1984) and for novel stimuli such as dot patterns (Gabrieli, Milberg, Keane, &

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Corkin, 1990). Priming also can occur independently of explicit remembering in normal populations. This has been shown both through functional dissociation, in which variables such as levels of processing (e.g., Jacoby, 1983) or study–test delay interval (Tulving, Schacter, & Stark, 1982) affect repetition priming and explicit remembering differently, and stochastic dissociation, in which performance on priming and performance on explicit memory tests are found to be statistically independent (e.g., Tulving et al., 1982; see Poldrack, 1996).

The Relationship Between Skill Learning and Priming

The findings that skill learning and priming both are spared in amnesia, and that both can be dissociated from explicit remembering in normal participants, lead to the question of whether these memory phenomena critically rely on, or share, a single underlying mechanism. In addition to both skill learning and priming being dissociable from explicit remembering in normal memory and amnesia, these two memory phenomena exhibit several other similarities that have led us (Cohen & Eichenbaum, 1993) and others (Logan, 1990) to suggest that these memory phenomena depend critically on a single memory mechanism. Perhaps most important is that both skill learning (Logan, 1988; Masson, 1986) and priming (Roediger, Weldon, & Challis, 1989; Tulving & Schacter, 1990) can result in inflexible memory representations, which require the reinstatement of the same processing operations that were engaged when the stimuli were first encountered (see Cohen & Eichenbaum, 1993) in order for maximal facilitation to be observed. In addition, it has also been noted that the decrease in reaction time (RT) for both skill learning (Newell & Rosenbloom, 1982) and priming (Logan, 1990) is characterized by a negatively accelerated power function.

Claims of Dissociation: Definitional

Although skill learning and priming share certain basic properties, there are also differences between the phenomena that have led to suggestions that they rely on different underlying mechanisms. On purely definitional grounds they differ in their stimulus specificity. Priming is tied to the specific stimuli that were encountered in the experiment (e.g., to the specific words read in mirror-reversed text). This differs from skill learning, which is defined as applying over potentially all stimuli in the practiced domain (e.g., to new examples of text presented in the same mirror-reversed format). Beyond this definitional difference, there are two recent lines of evidence, one involving neuropsychological studies and the other involving studies of normal populations, which have been cited as evidence for separate skill learning and priming mechanisms. We consider each of these lines of evidence in turn.

Claims of Dissociation: Neuropsychological Studies

Several neuropsychological findings have been cited as demonstrating dissociations between skill learning and priming. Martone, Butters, Payne, Becker, and Sax (1984) first examined this question by looking at the two memory phenomena in patients with Huntington's Disease (HD), amnesic patients, and normal control participants. Across testing days, blocks of mirror-reversed text were presented, with half of the stimuli appearing only once in the experiment (nonrepeated) and half of the stimuli appearing in each block (repeated). Martone et al. (1984) found that skill learning and priming dissociated in HD patients, who were mildly impaired at skill learning while exhibiting normal priming relative to the other participants. In a subsequent study of several participant groups, across two different tasks, Heindel, Salmon, Shults, Wallicke, and Butters (1989) found that HD patients exhibited impaired motor skill learning in a rotary pursuit task but normal lexical priming in a word-stem completion task (in line with the earlier finding), whereas patients with Alzheimer's disease (AD) exhibited the opposite pattern of performance (normal motor skill learning but impaired lexical priming).

These findings seemed to suggest a dissociation between skill learning and priming and led to claims that these memory phenomena rely critically on neurally distinct memory mechanisms. However, further consideration of these findings calls such claims into question. First, these studies required comparisons of performance across participant groups that were complicated (particularly in Martone et al., 1984) by differences in baseline performance across groups, making the multiple comparisons of skill learning and priming on different scales less than straightforward. There is a more fundamental concern about the Heindel et al. (1989) study, in which skill learning and priming were examined in different task domains (perceptual-motor performance vs. lexical processing). Rather than reflecting a dissociation between skill learning and priming, the results observed by Heindel et al. may actually reflect a dissociation between processing in perceptual-motor versus lexical domains. The fact that a patient group with motor problems (HD patients) showed impaired motor skill learning but normal lexical priming, and a patient group with language deficits (AD patients) showed an impairment in lexical priming but normal motor skill learning, may well say more about motor and lexical performance in these two patient populations than about the relationship between skill learning and priming generally.

The aforementioned concerns about comparing skill learning and priming across domains and across groups, and our conclusion that the claims of dissociation between skill learning and priming are overstated, have been borne out by subsequent research by other investigators. Keane, Gabrieli, Fennema, Growdon, and Corkin (1991) found that patients with AD can exhibit normal priming, for example, in a perceptual identification task. And both Gabrieli, Stebbins, Singh, Willingham, and Goetz (1997) and Willingham and Koroshetz (1993) have found that patients with HD can exhibit normal skill learning, even in certain motor tasks, such as mirror-tracing or learning a new stimulus-to-response mapping. Thus, the data do not support the notion that HD involves a selective deficit in skill learning while AD involves a selective deficit in priming. Moreover, when

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Deweer, Pillan, Michon, and Dubois (1993) examined skill learning and priming together in the mirror-reading task, they found that AD patients exhibited both skill learning and priming indistinguishable from that shown by normal controls, whereas the AD patients were severely impaired on tests of explicit remembering. Relatedly, a study that examined skill learning and priming in the mirror-reading task using functional magnetic resonance imaging found that common brain regions exhibited changes in activity related to both skill learning and priming (Poldrack, Desmond, Glover, & Gabrieli, 1998). These findings highlight the need to examine skill learning and priming within the same tasks and the same neuropsychological populations.

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Claims of Dissociation: Studies With Normal Populations

Work by Schwartz and Hashtroudi (1991) and by Kirsner and Speelman (Kirsner & Speelman, 1996; Kirsner, Speelman, & Schofield, 1993) with normal populations has explored certain empirical dissociations between skill learning and priming that might suggest different underlying mechanisms for these memory phenomena. Schwartz and Hashtroudi examined skill learning and priming in word identification, word fragment completion, and inverted text reading, with each task involving the presentation of both repeated and nonrepeated stimuli across blocks. They reported that in word identification the amount of priming was unrelated to pretest level of skill, and that priming and skill learning were affected differently by word frequency (a finding also reported by Kirsner & Speelman, 1996, for lexical decision). In addition, the amount of skill learning and amount of priming were uncorrelated across participants in both word identification and inverted-text reading tasks. Kirsner and Speelman's work in lexical decision indicated that priming did not continue to increase with practice, despite increasing level of (general) skill, with nearly all of the priming occurring on the first repetition. They interpreted their findings as constituting a failure to replicate Logan's (1990) report that priming and skill learning show the same power function speed-up.

Do these empirical dissociations between skill learning and priming demonstrate that the two memory phenomena are supported by independent memory systems? Dunn and Kirsner (1988), among others, have argued that tasks may exhibit dissociations, and even double dissociations, without requiring an explanation positing separate memory systems. What we really need to know is whether a single memory mechanism is capable of producing the various relationships between skill learning and priming observed empirically. In the section entitled *Computational Analysis of Skill Learning and Priming,* we present simulation results from two different single-mechanism computational models (connectionist and instance models) showing that the variable relationships between skill learning and priming observed empirically, including lack of correlation, can indeed be obtained within a single memory mechanism. The results of this modeling effort undermine claims that the empirical dissociations discussed above require positing multiple memory mechanisms.

Few experiments have attempted to assess skill learning and priming simultaneously; in order to do so, some stimuli are presented only once (nonrepeated) and other stimuli are presented repeatedly. In such a design, skill learning is reflected in improvements in performance across trials for the nonrepeated stimuli, and priming is reflected in facilitation of performance for repeated stimuli compared to nonrepeated stimuli at the same point in training (see Cohen & Squire, 1980).

Logan's (1990) exploration of priming differed from the above by having a single set of stimuli repeated multiple times to examine priming across repetitions. Priming was measured as the reaction time speed-up for the *n*th repetition of repeating stimuli relative to the first presentation, and these response times decreased as a power function of repetitions. Unfortunately, this particular approach provides no way to discern how much of the performance improvement curve is item-specific (i.e., priming) and how much is item-general (i.e., skill learning). Over the course of these experiments, performance improvements might also be expected to occur for new exemplars of stimuli (nonrepeated stimuli) within that domain. Without examining the difference between performance on repeated stimuli and new stimuli at various points during the course of the experiment, item-specific (i.e., priming) and item-general (i.e., skill learning) effects cannot be distinguished.

Kirsner and Speelman (1996) presented their participants with some lexical decision stimuli that appeared repeatedly throughout the experiment and other stimuli that appeared only once during the experiment. This permitted them to observe both skill learning and priming, and thereby to evaluate the relationship between these two memory phenomena. They found that priming, after reaching a maximum amount early in acquisition, remained unchanged despite additional improvements in general skill, in contrast to Logan's (1990) report of the same power function speed-up for priming as for skill learning. This finding is modeled conceptually using connectionist and instance models below, the results of which demonstrate that this apparent dissociation does not imply distinct skill learning and priming mechanisms.

The Utility of the Skill Learning–Priming Distinction

We have argued above that the only way to study the relationship between skill learning and priming is to assess both memory phenomena within the same task, and we have attempted to illustrate the pitfalls of alternative empirical strategies. In the studies reported below, both repeating and nonrepeating stimuli were presented to participants across a series of blocks, which permitted us to explore both item-specific and item-general performance changes and the relationship among them. We have also argued above that various empirical relationships between skill learning and priming, including independence of the two memory phenomena, can be modeled in single memory mechanism models, which we demonstrate in the section of the General Discus-

sion entitled *Computational Analysis of Skill Learning and Priming*. The results of this computational analysis raise some important issues about the possible relationships between skill learning and priming, and about their implications for understanding the nature and organization of memory. Accordingly, before proceeding to a description of the present studies, we discuss certain issues concerning how to think about and how to study the relationship between skill learning and priming.

Repetitions at Differing Levels of Skill

One persistent problem in the literature is the conflation of level of (item-general) skill with the effects of (item-specific) repetition. Even the studies that have been careful to include both repeating and nonrepeating stimuli in order to permit assessment of skill learning and priming simultaneously have failed to gain a clean assessment of the amount of priming observed for newly repeating items at different levels of skill. How does the amount of priming for an item first introduced at a given level of skill relate to the level of skill achieved? Is the priming observed when already expert in the domain the same as the amount of priming observed when just a beginner in that domain? The only way to answer this question would be to present—and to assess priming for—different sets of repeating items that begin repeating at different levels of skill, rather than to have a single set of repeated items appear throughout the experiment. No previous work in the literature, to our knowledge, has implemented this approach; it is the approach we take in the present studies.

This methodological issue is particularly important in thinking about the relationship between skill learning and priming because it is just this issue—the level of skill of the participant—that distinguishes between and, indeed, serves to separate the skill learning and priming literatures. Priming is almost invariably assessed in tasks in which participants are already expert—word reading, picture naming, and so forth. Skill learning, by contrast, is rarely if ever studied in such domains because participants have so much expertise already, making performance improvements within a short session difficult to discern or at least difficult to measure accurately. Most studies of skill learning, accordingly, involve domains in which participants are initially novices and can show precipitous gains in performance across nonrepeating stimuli. Even in the few cases where priming is assessed for items with no prior representation in memory (e.g., nonwords or dot patterns), and hence participants can plausibly be called nonexperts, the paradigm involves a simple study-then-test procedure in which little general skill learning is likely to occur. Given how separate the procedures end up being for the study of skill learning versus the study of priming, it should come as no surprise that the literatures have remained largely separate and that there are not already-existing paradigms for effectively studying the two phenomena simultaneously. However, by designing paradigms specifically to assess the amount of priming for materials that are introduced at differing levels of skill, from novice to increasingly expert, the present studies can bridge this gap between literatures and can potentially offer a richer understanding of the relationship between skill learning and priming.

What Is an "Item"?

In defining skill learning and priming, we have discussed the idea of item-general and item-specific performance changes. But, what exactly constitutes an "item"? Intuitively, it often seems clear what an item is for a given task. For example, words would seem to be the items in single-word processing tasks, such as speeded single-word reading, perceptual identification of (single) masked words, or lexical decision. Note, however, that such intuitions seem really to be about what we discern as the fundamental unit of processing in a given task, whether or not that unit of processing is also the locus of memory effects. Consider, for example, work by Rueckl (1995) and by Dorfman (1994) demonstrating the priming of sublexical features of words in a word-fragment completion task. Thus, although the experimenter and participant might see the task as being about words, and words might indeed be the basic unit of processing in that task, the locus (or one of the loci) of memory is at level(s) of the word processing system other than (or in addition to) the whole-word level. In the cascade of information processing that underlies word recognition, memory can exert its effects not only at the whole-word level but at other ("lower" or "earlier") levels as well. (For similar conclusions about the loci of memory in artificial grammar learning, see Perruchet & Pacteau, 1990, and Gomez & Schvaneveldt, 1994).

Any attempt to explore the relationship (or distinction) between skill learning and priming must depend on defining what an item is and on identifying the point(s) in the cascade of information processing at which memory can exert its effects in the task of interest. As we have discussed elsewhere (Cohen, Poldrack, & Eichenbaum, 1997), memory can exert its effects at various levels in a given system, and what an item is can be defined differently at each level. When reading words, for example, there are levels of processing that treat line orientations as the items; other levels that treat letters as the items; others that might treat letter combinations, such as syllables, as the items; and a level that treats whole words as items. Each of these levels, separately and in combination, might be fully capable of expressing memory that could support priming or skill learning (Cohen, 1984; Cohen & Eichenbaum, 1993). The present studies are designed in such a way as to permit performance changes tied to one or another of several different levels of processing to be assessed, each of the different levels presumably treating different aspects (or subsets) of the stimuli as items.

The Present Studies

The present studies involved a digit entering task (DET) in which participants were presented with multidigit number strings on a computer monitor and entered them into a computer keypad (similar to a task used in Fendrich, Healy,

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& Bourne, 1991). Participants in effect perform a succession of choice RT trials, one for each of the digits in the string. The DET used in the present studies bears much similarity to what in the literature is referred to as sequential reaction time (SRT) tasks, but has a number of novel features designed to permit us to address the issues raised in the preceding discussion about the relationship between skill learning and priming.

The SRT Task

In a typical SRT experiment (e.g., Nissen & Bullemer, 1987), choice RT trials are presented in succession, and the sequence of targets follows a pattern defined by the experimenter. To the extent that information is learned about the patterned sequence of targets, response times on trials with that patterned sequence are faster than response times on trials with a random sequence. Normal participants and amnesic patients both show increasingly facilitated performance for the patterned sequence versus a random one (Nissen & Bullemer, 1987; Reber & Squire, 1994).

This speedup for the patterned sequence in the SRT has generally been referred to as skill learning, perhaps because of its seemingly motoric nature (though see Keele, Jennings, Jones, Caulton, & Cohen, 1995, and Willingham, Nissen, & Bullemer, 1989, for evidence that learning in this task relies on nonmotoric components). However, on the definitions of skill learning and priming advocated above, performance facilitation in the SRT task seems to fall into the class of phenomena more accurately characterized as repetition priming (cf. Curran, 1997). The performance improvement observed on this task occurs for a single repeating sequence as compared to a novel sequence, just as it does in various other examples of priming.

The DET

By contrast, the DET used in the present work has instituted a set of procedural changes from previous studies that permitted us to examine the relationship between skill learning and priming and to better understand the nature of what is being learned, in terms of the loci of the memory effects. First, we presented both repeating and nonrepeating digit strings, permitting assessment of both skill learning and priming simultaneously. Because there is a large number of repeated strings, compared to the single 6- to 10-element long pattern that repeats in most SRT studies, performance in the DET may suffer less from intrusion by explicit remembering of the stimuli.

Second, we introduced new repeating digit strings at multiple points during the development of general skill in the task. This permitted assessment of the extent to which priming is affected by the level of skill or expertise in the domain. Participants were likely to begin our studies as relative novices, having had limited experience in entering multidigit strings into a computer keypad.

Third, the stimuli were constructed so as to have structure at several different levels in the processing chain; learning across trials could plausibly occur at any or all of the levels. Single digits could occur in each position within a digit string, but only certain digit-to-digit transitions appeared (repeatedly) whereas others never appeared, and only certain whole digit strings repeated whereas others did not. The extent to which structure at each of these levels influences digit entering performance, both in terms of response time and in terms of the nature of errors produced (analyzed in the manner of Grudin, 1983, for transcription typing), was assessed in the present studies.

General Method

Similar methods and procedures were used for each of the experiments reported here.

Procedure

Five-digit strings (e.g., 83972) were presented individually to participants in white on a computer screen, with instructions to "enter" the digit strings using a numeric keypad. The onset of each digit string was preceded by a 500-ms fixation cross. No visual feedback was provided for the response. After entering the 5-digit string, the participant pressed the "enter" key, which triggered the next trial after a 1,000-ms intertrial interval. Participants were instructed not to correct errors; they were told to press the "enter" key if they made an error, in order to proceed to the next trial. In each session, some strings appeared multiple times (repeated), whereas other strings appeared only once (unique).

Stimulus presentation was controlled by a Dell 210 microcomputer with a Dell VGA monitor. Each 5-digit string measured approximately 7 mm × 17 mm on the monitor. The response time for each keypress was measured to the nearest millisecond, and the accuracy of each keypress was recorded. Participants were tested individually in a testing room and were provided with wrist pads for comfort.

Stimuli

Training stimuli were chosen from a subset of all possible five-digit strings on the basis of the following three constraints: (a) Only digits 1 through 9 appeared in the digit strings (the digit 0 was excluded); (b) digits never repeated in consecutive positions within a digit string (e.g., 44 never appeared as part of a string); (c) digit strings were constrained to exhibit certain first-order transitions between digits. Of the eight possible first-order transitions from one digit to another (after excluding the digit 0 and immediate repetitions of any digits), four were chosen at random, and only strings following this set of transitions were included as stimuli for that transition set. See Figure 1 for a diagram of the rule sets and examples of the stimuli produced using these transition rules.

Half of the participants in each experiment were presented with digit strings created from one transition rule set, and the other half were presented with strings created from the complementary rule set.

For each participant, digit strings were randomly assigned to various experimental conditions and then tested to equate the statistical structure of stimuli across conditions in terms of the probability of each digit occurring in each position and the probability of each first-order transition between digits (regardless of position). This was accomplished for each participant by choosing string sets at random and then testing the resulting digit position and transition probability distributions of the sets for the different conditions, using a chi-square test. This choose-then-test

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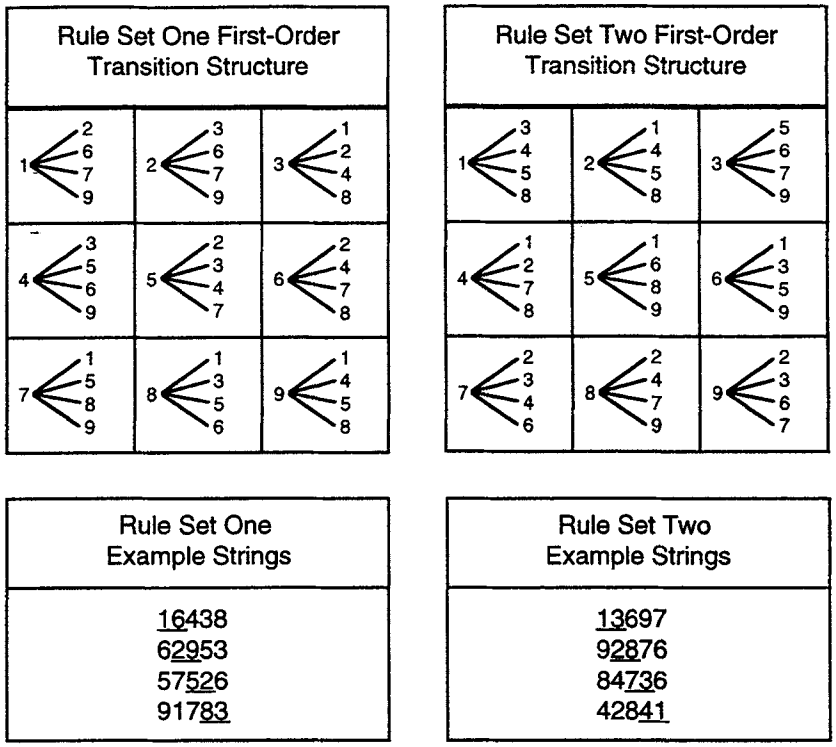


Figure 1. Diagram of two complementary first-order transition rule sets used in the present experiments and example stimuli.

procedure was performed 10,000 times by computer, and the best-matching set of strings was assigned for the given participant.

In each experiment, participants were transferred at some point in training (different for the various conditions) to strings constructed from a new (transfer) transition rule set that was complementary to the acquisition rule set (see Figure 1). These transfer strings were equated with the training strings with respect to probability of appearance of each digit in each position, using the selection method described above; thus, any observed effects of transfer could not be attributed to possible zero-order statistical differences between the training and transfer sets of digit strings.

Data Analysis

Response times were separated into two components, latency and interkeystroke interval (IKI), as is common in the study of transcription typing performance (e.g., Sternberg, Monsell, Knoll, & Wright, 1978). Latency was the time from presentation of the stimulus to the first keypress (for the first digit of the string). IKI was the average interval between each subsequent keystroke (for digits 2–5); these IKIs should not be interpreted as response times in a strict sense because one cannot know with certainty when the processing began that led to the response. If any of the digits of a given string was entered incorrectly, an error was recorded for that stimulus in analyses of accuracy and none of the associated keypress times were included in analyses of response times. Analysis of accuracy and error types is further described in Appendix A.

Priming was measured proportionally in Experiments 1 and 2 by dividing the difference in response times between repeated and unique items by the response time for unique items from the first block of training. This measure was used to allow comparison of priming over different experiments in which the baseline response times differed and to reduce variability of priming scores between participants. Absolute priming scores (in milliseconds) are reported for Experiment 3 to allow fitting of power functions in a manner comparable to previous studies (Kirsner & Speelman, 1996).

The significance level for all statistical tests was $\alpha = .05$. Unless otherwise noted, all analyses of variance (ANOVAs) used repeated measures analysis with the Huyhn-Feldt correction for nonsphericity (Hays, 1988). Response time and overall accuracy data were averaged over three successive blocks for graphical presentation, but the blocks were treated individually for all statistical analyses.

Experiment 1

Experiment 1 examined skill learning and priming in the DET over three sessions of training. This experiment was conducted to answer two primary questions. First, how does the amount of priming vary with the level of overall skill? This question was answered by comparing priming over subsequent blocks of training for items that begin repeating in different sessions. Second, are the first-order statistical regularities (i.e., sequential or transition structure) of the stimuli represented as part of the knowledge supporting skill acquisition? This question was addressed in two ways. Performance on new items from the training transition rule set was compared with performance on new items from the complementary (transfer) transition rule set introduced during a transfer block. To the extent that performance on the complementary (transfer) rule set items was slower, or less

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accurate, or both, statistical regularities were assumed to underlie some portion of the learning in the DET. Additionally, participant responses on error trials were analyzed for evidence of knowledge about the transition structure. To the extent that errors frequently resulted in strings that substituted digits incorporating permissible transitions, in line with the acquisition transition structure for that participant, there would be evidence of knowledge about the first-order sequential regularities.

Method

Participants

Sixteen undergraduate students from the University of Illinois at Urbana-Champaign participated in the experiment. They received class credit or were paid $15 for their participation in three 1-hr sessions, each separated by 1 day of rest. All participants were right-handed.

Procedure

Participants performed digit entering as described in the General Method section for three sessions. Each session consisted of 12 blocks of 48 trials, for a total of 576 trials. On each trial, a string was presented and remained visible until the participant entered the whole string and pressed the enter key to proceed to the next trial. Participants received a short break after every 72 trials.

Stimuli

Stimuli were chosen as described in the General Method section. Table 1 presents the stimulus makeup for a single block in each session of Experiment 1; other than the "new rule unique" strings, all strings were created using the acquisition rule set for each particular participant. In Session 1, the "Session 1 repeat" (S1R) strings were first presented in Block 1 of Session 1 and were presented again in each subsequent block of the experiment. In the first block of Session 2, new to-be-repeated (i.e., "Session 2 repeat" [S2R]) strings were introduced, and then presented again in each subsequent block of the experiment. In these first two sessions, nonrepeating strings (uniques) were also presented in each block. In Session 3, nonrepeating strings generated from the complementary transition rule set (new rule uniques) were also presented, in addition to the other categories of stimuli. Different numbers of unique strings appeared in different sessions; this was done to keep the average repetition lag constant at 48 for repeating strings across sessions.

Table 1
Stimulus Makeup for Each Block in Each Session of Experiment 1

Session and trial type	Rule set[a]	No. per block
1		
Session 1 repeat (S1R)	1	12
Acquisition rule unique (unique)	1	36
2		
S1R	1	12
Session 2 repeat (S2R)	1	12
Unique	1	24
3		
S1R	1	12
S2R	1	12
Unique	1	12
New rule unique	2	12

[a]Half of the participants were presented acquisition stimuli generated from Transition Rule Set 1, and the other half of the participants were presented acquisition stimuli generated from Transition Rule Set 2.

Results

Response time data (latency and IKI) for Experiment 1 are presented in Figure 2.

Latency

Skill learning. Significant skill learning occurred for latency, evidenced by a decrease in latencies over blocks for unique strings, linear trend $F(1, 15) = 181.93$, $MSE = 14{,}782.8$.

Priming. Significant priming was found for response latencies; proportional priming data are shown in Figure 3 for S1R and S2R items, with the results of planned comparisons between repeat and unique strings shown in the left two columns of Table 2. Significant priming for S1R strings was observed in Sessions 2 and 3. Trends in the direction of priming were observed for S2R strings but did not reach significance. These results suggest that priming is weaker when repetition begins later in training, but a separate 2 (item type: S1R vs. S2R) × 24 (repetitions) ANOVA on proportional priming scores revealed no significant effects ($ps > .38$), suggesting that the difference between S1R and S2R priming was small. S1R priming, when analyzed alone, demonstrated a significant linear increase, $F(1, 15) = 5.37$, $MSE = 0.005$, whereas for S2R items the linear trend was not significant ($F < 1$). These data indicate that priming grew across repetitions for strings introduced early in training, but not for strings introduced later in training.

Transition rule effects. The effects of transferring to a new set of transition rules in Session 3—that is, of introducing unique digit strings constructed from the complementary rule set (new rule uniques)—were examined using a 4 (trial type: S1R, S2R, unique, new rule unique) × 12 (blocks) ANOVA. There was a main effect of trial type, $F(3, 45) = 15.27$, $MSE = 29{,}143.8$. Planned comparisons showed that latency for new-rule unique digit strings ($M = 1{,}189.56$ ms) was significantly slower than for unique digit strings ($M = 1{,}121.06$ ms), $F(1, 15) = 15.45$, indicating that changes in digit-to-digit transition structure interfered with performance. Latency for new rule unique items in Session 3 was faster than for unique items in Session 1, $F(1, 15) = 14.33$, $MSE = 86{,}768$, suggesting general improvement of performance related to the entry of individual digits (i.e., zero-order learning).

IKI

Skill learning. Significant skill learning was observed for IKI, as seen in decreasing IKI over blocks, linear trend $F(1, 15) = 260.0$, $MSE = 856.5$.

Priming. Priming was also observed for IKI; these effects are shown in Figure 3. The results of planned

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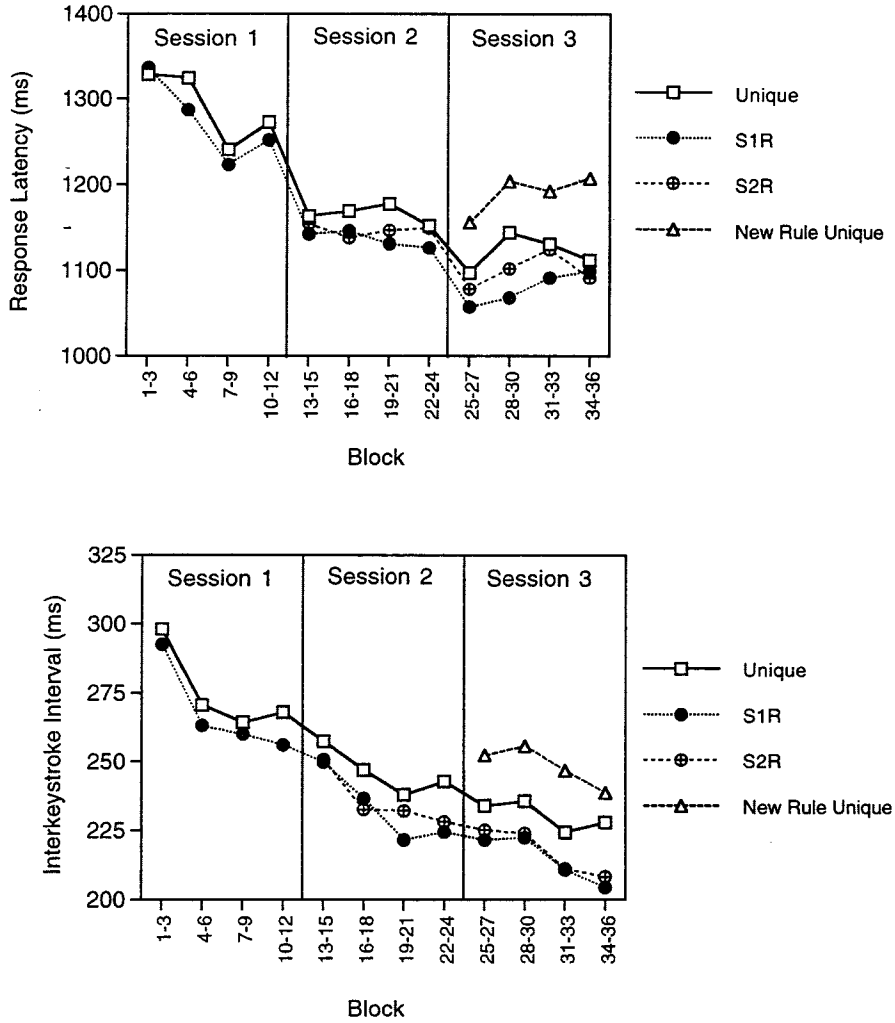


Figure 2. Digit entering response latencies and interkeystroke intervals from Experiment 1 across blocks for each of four stimulus types. S1R and S2R indicate repeating stimuli introduced during Sessions 1 and 2, respectively. Unique and New Rule Unique refer to nonrepeating stimuli generated from the same and different rule sets, respectively, used to generate the repeating stimuli. Note that New Rule Unique stimuli were presented only during Session 3. For data presentation only responses were collapsed over groups of three blocks.

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comparisons between repeating and unique strings are shown in the right two columns of Table 2. For IKI, unlike for latency, significant priming was observed for both S1R and S2R strings in each session. Priming for the first 24 repetitions of S1R and S2R strings was compared using a 2 (repeat item type: S1R vs. S2R) × 24 (repetitions) ANOVA on proportional priming scores. There was no significant difference between priming for S1R and S2R strings, and no significant interaction ($Fs < 1$). S1R priming demonstrated a significant linear increase, $F(1, 15) = 35.60$, $MSE = 0.0003$, as did S2R priming, $F(1, 15) = 8.79$, $MSE = 0.0003$. These data suggest that the effects of repetition on IKI were incremental and did not vary depending on the point at which new repeating strings were introduced during training.

Transition rule effects. The effect of rule transfer (i.e., the introduction of new rule uniques) on IKI was examined using a 4 (trial type: S1R, S2R, unique, new rule unique) × 12 (blocks) ANOVA. There was a main effect of trial type, $F(3, 45) = 45.96$, $MSE = 987.7$. Planned comparisons showed that IKI for new rule unique digit strings ($M = 248.28$ ms) was significantly slower than for unique strings ($M = 230.43$ ms), $F(1, 15) = 30.96$, again indicating that changes in digit-to-digit transition structure of the strings hindered performance. IKI for new rule unique items was faster than performance on unique items in Session 1, $F(1, 15) = 16.60$, $MSE = 3{,}174$, indicating general improvement with digit-entering practice.

Relationship Between Skill Learning and Priming

The correlation between skill learning and repetition priming in Experiment 1 was examined by comparing skill

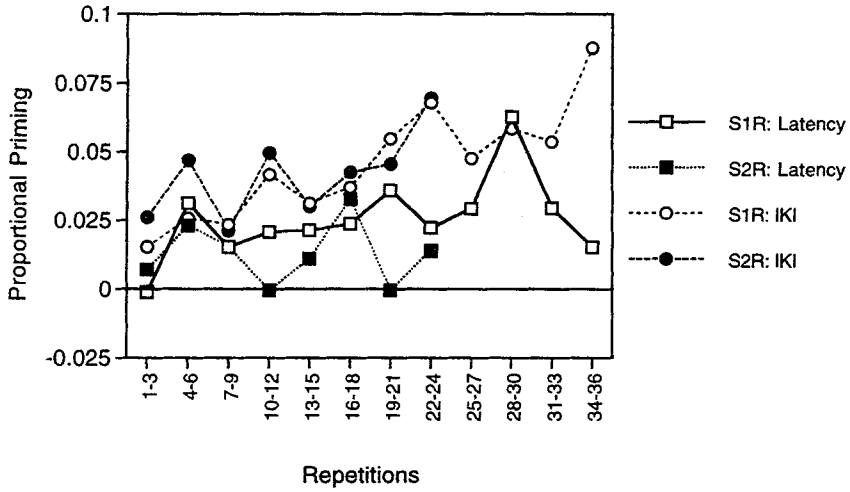


Figure 3. Proportional priming for latency and interkeystroke interval (IKI) from Experiment 1 across blocks for Session 1 repeat (S1R) and Session 2 repeat (S2R) stimuli. Priming was computed as [(unique RT_n − repeat RT_n)/unique RT_1], where subscripts indicate block number. Thus, priming difference scores were scaled to Block 1 unique reaction times (RTs). For data presentation only, responses were collapsed over groups of three blocks.

learning scores (the difference between unique response times over the first three blocks of Session 1 and the last three blocks of Session 3) and priming scores (the difference between S1 repeat and unique items over the last three blocks of each session). These correlations are presented in Table 3. A significant negative correlation was obtained between skill learning and S1R priming in Session 3 for the latency measure, and a marginally significant negative correlation was obtained between skill learning and S2R priming in Session 3; the correlation between skill learning and priming in Sessions 1 and 2 was negative but not significant. For IKI, no correlations between skill learning and priming were significant. Thus, the relationship between skill learning and repetition priming varied with the measure of performance.

Table 2
Results of Planned Comparisons for Priming in Experiments 1 and 2

Session	Comparison	Latency $F(1, 15)$	Latency *MSE*	IKI $F(1, 15)$	IKI *MSE*
		Experiment 1			
1	S1R vs. unique	1.79	15,410.9	8.31*	617.7
2	S1R vs. unique	5.63*	14,146.5	22.32*	742.5
3	S1R vs. unique	5.86*	29,143.8	23.74*	987.7
2	S2R vs. unique	2.25	14,146.5	13.92*	742.5
3	S2R vs. unique	1.61	29,143.8	17.23*	987.7
		Experiment 2			
1	S1R vs. unique	5.45*	14,492.3	6.75*	607.2
2	S1R vs. unique	8.19*	60,594.6	36.19*	476.3
3	S1R vs. unique	4.92*	27,051.3	30.08*	353.8
2	S2R vs. unique	1.45	60,594.6	2.66	476.3
3	S2R vs. unique	2.75	27,051.3	10.49*	353.8

Note. IKI = interkeystroke interval; S1R = Session 1 repeat; S2R = Session 2 repeat.
*$p < .05$.

Accuracy and Error Analysis

The accuracy and error analyses are described fully in Appendix A. Analysis of errors confirmed the findings of the response time analyses; accuracy was greater for S1R than

Table 3
Correlations Between Skill Learning and Repetition Priming for Experiments 1 and 2

Session	Comparison	Latency *r*	Latency *p*	IKI *r*	IKI *p*
		Experiment 1			
1	S1R	−.363	.17	.202	.46
2	S1R	−.253	.35	−.389	.14
3	S1R	−.807	<.0001	−.212	.43
2	S2R	−.334	.21	−.107	.70
3	S2R	−.495	.05	.229	.40
		Experiment 2			
1	S1R	.160	.56	.276	.31
2	S1R	.154	.58	.089	.75
3	S1R	−.113	.69	.224	.41
2	S2R	.359	.18	.276	.31
3	S2R	.036	.90	.167	.54

Note. Skill learning was defined for all comparisons as the difference in response times for unique items on the first three and last three blocks of the experiment; priming was defined as the difference in response times between unique and repeated items in the last three blocks of the appropriate session. Exact *p* values were derived using Fisher's *r*-to-*z* transformation. IKI = interkeystroke interval; S1R = Session 1 repeat; S2R = Session 2 repeat.

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for unique items in Sessions 2 and 3 and was reduced for new rule items in Session 3. No linear trends were noted for accuracy on S1R items over all three sessions, but a significant linear decrease in accuracy was evident for unique items across all three sessions. These data suggest that the decrease in unique response time may have been associated with a speed–accuracy tradeoff, but differences between conditions cannot be attributed to speed–accuracy tradeoffs. The primary finding of the analysis of error types was a predominance of errors following the transition rule set from which items were created. This finding provides additional evidence for the learning of transition rule information.

Discussion

Experiment 1 demonstrated both skill learning and priming in the DET, permitting us to make some observations about the relationship of priming to level of skill. The relationship differed as a function of whether the measurement of interest was mean latency to initiate responding or mean interval between the remaining keystrokes (IKI). For IKI, strong priming was observed for all sessions, regardless of when repeating digit strings were introduced; there were no significant differences between priming for repeating digit strings introduced in Session 1 versus Session 2. For latency, however, priming was observed only for digit strings that began repeating in Session 1, and only reached significant levels in Session 2 after these digit strings had been seen 12 times. Repeating digit strings introduced in Session 2 did not exhibit the increase in priming seen for digit strings introduced in Session 1. Skill learning and priming were negatively correlated later in training for latency but not for IKI, suggesting that these two measures may reflect the effects of memory at different points in processing.

The difference in results for latency and IKI measures suggests that IKI may be a more powerful measure of memory for the effects of repetition in this domain. The fact that the priming in the DET is particularly robust for IKI seems consistent with an interpretation that perceptual-motor priming occurs primarily through the facilitation of the articulatory processes involved in implementing the sequential movements in the task (see Rumelhart & Norman, 1982, for a model of coordinating typing movements), rather than in the encoding of digit strings or the planning and programming of a response in advance of initiating movement.

The other major result from Experiment 1 relevant to understanding the relationship between skill learning and priming is the finding that there were at least three distinct levels of organization at which memory was manifested in this task. A significant portion of the learning effect in the DET was tied to the specific digit-to-digit transitions that appeared across the digit strings as specified by the first-order transition rules. Evidence for learning of transitions comes from the significant decrement in performance caused by the introduction of unique digit strings constructed from the complementary rule set (new rule uniques). The error data corroborate this finding; the largest category of errors was transition-legal errors, involving substitutions of the presented digit string with one or more digits that were consistent with the first-order transition rules. However, note that transferring to unique digit strings constructed from the complementary rule set did not eliminate all of the performance gains acquired across blocks. Performance on new rule uniques was still superior to initial performance on the original uniques, indicating that a portion of the learning was common to both sets. The information common to both sets involves knowledge about single digits, namely the spatial or motoric mapping of the individual digits to the numeric keypad. Thus, there were both one-digit and two-digit (digit-to-digit transition) effects evident in this experiment. The priming effects in Experiment 1 demonstrate a third level of organization in memory, namely that of higher order transition knowledge such as second-order, third-order, or fourth-order (whole-string) knowledge. The existence of these three distinct levels of memory effects, corresponding to the three levels of stimulus structure built into the task, calls into question the utility of a binary distinction between skill learning (i.e., zero-order knowledge) and repetition priming (i.e., whole-string knowledge).

Experiment 2

The priming evident in Experiment 1 was relatively modest, increasing only weakly over multiple repetitions. Experiment 2 incorporated a modification of the DET procedure designed to enhance whole-string priming effects. Participants were encouraged to encode the digit strings as whole 5-digit items, which is the level of stimulus structure at which priming is assessed, rather than encoding them as sets of single digits or digit-to-digit transitions. This was accomplished in the present experiment by removing the stimulus from the screen as soon as the participant entered the first digit of the 5-digit string. In the previous experiment, by contrast, it was possible for participants to encode only a portion of the string, then program and execute the appropriate response, before encoding the rest of the digit string. The present experiment makes this strategy ineffective. An additional modification of the procedure was the addition of a test of recognition memory, administered in Session 3, to examine the level of explicit remembering of these stimuli.

Method

Participants

Sixteen undergraduate students from the University of Illinois at Urbana-Champaign participated in the experiment. They were paid $15 for their participation in three 1-hr sessions, each separated by 1 day of rest. All participants were right-handed.

Procedure

The procedure was the same as in Experiment 1, with two exceptions. First, the digit strings were removed from the display as soon as the participant made the first keypress. Second, Session 3 had only four blocks of digit entering trials (as opposed to 12 blocks in Experiment 1), followed by a recognition test of whole

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Table 4
Stimuli Used in the Recognition Task in Experiment 2, Along With Recognition Data From Experiment 2

Session[a]	Stimulus type	Number	Mean confidence rating	F (1, 14)[b]	SE
		Targets			
1	Repeat (S1R)	12	3.94*	33.23*	.08
1	Acquisition rule unique	12	3.02	0.006	.11
2	Repeat (S2R)	12	3.83*	25.94*	.15
2	Acquisition rule unique	12	3.41*	5.91*	.17
3	Acquisition rule unique	12	3.29*	18.98*	.16
3	New rule unique	12	2.61*	5.77*	.15
		Distractors			
	Acquisition rule	36	3.03	25.69	.12
	New rule	36	2.23		.14

Note. S1R = Session 1 repeat; S2R = Session 2 repeat.
[a]Session number indicates the session in which target stimuli originally occurred. [b]For repeat and acquisition rule unique stimuli, this F value represents test for difference from acquisition rule distractors; for new rule unique stimuli and acquisition rule distractors, this F value represents test for difference from new rule distractors.
*$p < .05$.

5-digit strings. Table 4 presents the various categories of target and distractor strings used in the recognition task.

Participants were shown each digit string individually and were asked to judge whether they had seen the string before, using a scale of 1 (*I am certain I didn't see it*) to 6 (*I am certain I saw it*). Participants were not allowed to enter the string into the keypad before making a judgment. Judgments were indicated by pressing one of the number keys at the top of the keyboard rather than on the numeric keypad used for DET trials. This procedure was used to lessen the likelihood that recognition judgments would be based on motoric "fluency" (e.g., Fendrich et al., 1991).

Stimuli

The stimuli for digit entering trials were created as in Experiment 1.

Results

Response time data, for latency and for IKI, are presented in Figure 4.

Latency

Skill learning. Significant skill learning was observed for latencies in Experiment 2, linear trend $F(1, 15) = 152.65$, $MSE = 57{,}692.5$.

Priming. Significant priming was observed for latency. Proportional priming, for both S1R and S2R items, is shown in Figure 5, with results of planned comparisons shown in the left panel of Table 2. The analyses demonstrated a pattern similar to that seen in Experiment 1: Priming for S1R strings was significant and grew over time, whereas priming for S2R strings was not significant (although once again the trends were in the right direction, and the effect seemed to increase slightly). As was observed in Experiment 1, a 2 (S1R vs. S2R) × 24 (repetitions) ANOVA failed to reveal a significant difference between S1R and S2R strings or any interaction with repetitions ($Fs < 1$). Linear trends in priming were evident for both S1R items, $F(1, 15) = 7.97$, $MSE = 0.004$, and S2R items, $F(1, 15) = 4.64$, MSE = 0.004.

Transition rule effects. Transition rule effects were analyzed using a 4 (blocks) × 4 (item type: S1R, S2R, unique, new rule unique) ANOVA on latencies. The main effect of trial type was significant, $F(3, 45) = 11.12$, $MSE = 27{,}051.3$. Planned comparisons revealed a significant difference between unique strings and new rule unique strings, $F(1, 15) = 9.18$; latencies for new rule unique strings ($M = 1{,}461.3$ ms) were significantly slower than for unique strings ($M = 1{,}373.2$ ms), indicating that changes in digit-to-digit transition structure interfered with performance. Performance on new rule unique items was faster than performance on unique items in Session 1, $F(1, 15) = 8.34$, $MSE = 438{,}885$, indicating general improvement with practice.

IKI

Skill learning. Skill learning was observed for IKI in Experiment 2, as it was for latency, linear trend $F(1, 15) = 132.5$, $MSE = 378.7$.

Priming. Significant priming was also observed for IKI in Experiment 2. Proportional priming scores are shown in Figure 5, and results of planned comparisons are shown on the right side of Table 2. Large priming effects were observed for S1R strings, which grew over subsequent blocks. For S2R strings, priming was not significant in Session 2 but became significant in Session 3. These effects seemed to grow in a manner qualitatively similar to S1R items. Priming for the first 24 repetitions of S1R and S2R strings was compared using a 2 (repeating item type: S1R vs. S2R) × 24 (repetitions) ANOVA on proportional priming scores. There was no significant difference between S1R and S2R strings, and no interaction ($Fs < 1$). Significant linear increases in priming were observed for S1R items, $F(1,$

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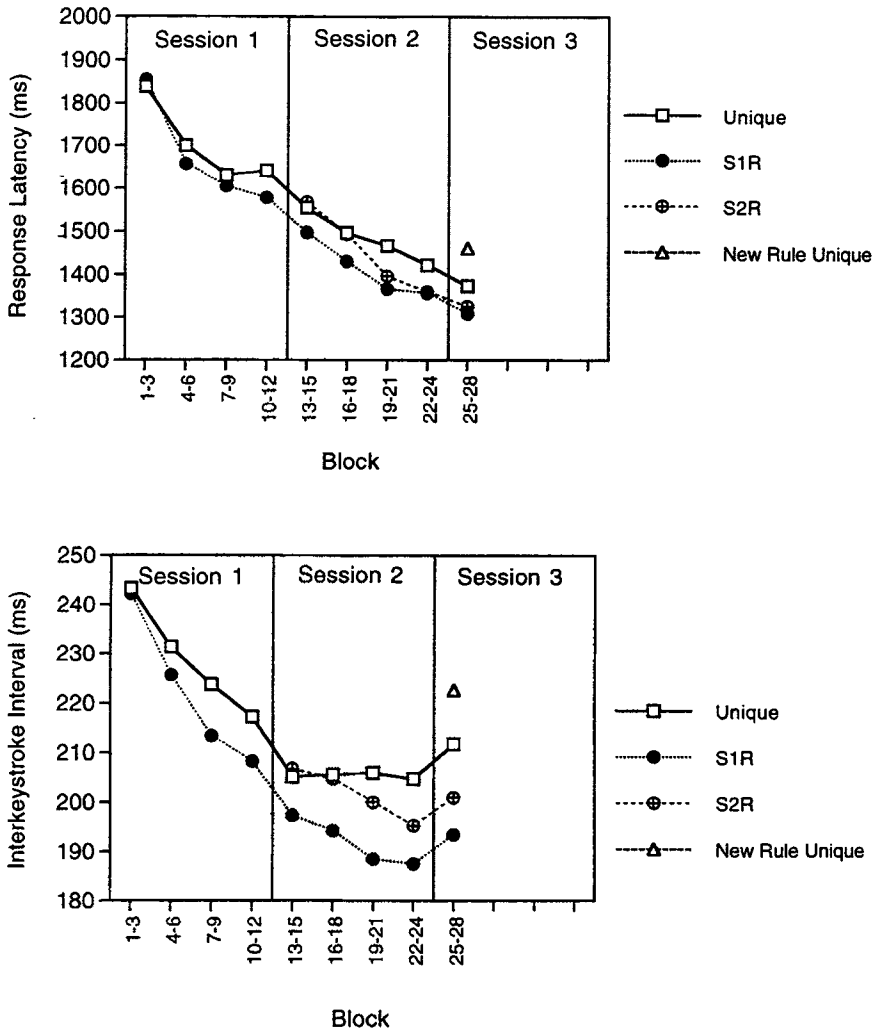


Figure 4. Digit entering response latencies and interkeystroke intervals (IKIs) from Experiment 2 across blocks for each of four stimulus types. S1R = Session 1 repeat; S2R = Session 2 repeat.

15) = 53.80, MSE = 0.003, and S2R items, $F(1, 15)$ = 22.67, MSE = 0.004.

Transition rule effects. The effects of transfer to new transition rules in Session 3 were examined using a 4 (blocks) × 4 (item type: S1R, S2R, unique, new rule unique) ANOVA. The main effect of item type was significant, $F(3, 45)$ = 29.33, MSE = 353.8. Planned comparisons showed that IKIs for new rule unique strings (M = 222.59 ms) were slower than IKIs for unique strings (M = 211.66 ms), $F(1, 15)$ = 10.81, indicating that changes in digit-to-digit transition structure interfered with performance. Performance on new rule unique items was faster than performance on unique items in Session 1, $F(1, 15)$ = 8.15, MSE = 1,293.

IKI for unique strings increased slightly from the end of Session 2 (M = 205.2 ms) to the beginning of Session 3 (M = 211.7 ms), suggesting that the introduction of new rule items may have resulted in interference on items from the acquisition rule set. However, a test for this increase was not significant, $F(1, 15) = 1.29, p > .27$. This issue is addressed directly in Experiment 3.

Accuracy and Error Analysis

Analyses of accuracy and errors are presented in Appendix A. The results of the error analysis parallel those of Experiment 1; accuracy on repeated items was greater than that for unique items, and accuracy on new rule unique strings in Session 3 was less than accuracy on acquisition rule uniques (though not significantly). No significant linear trends were found, suggesting that the response time findings were not compromised by speed–accuracy tradeoffs. The error analysis demonstrated, as in Experiment 1, that transition-legal errors predominated over other types of errors.

Recognition

Mean recognition ratings (on a scale of 1 to 6) for the various categories of target and distractor items are displayed in Table 4, including the results of planned comparisons between targets and distractors (recognition data from 1

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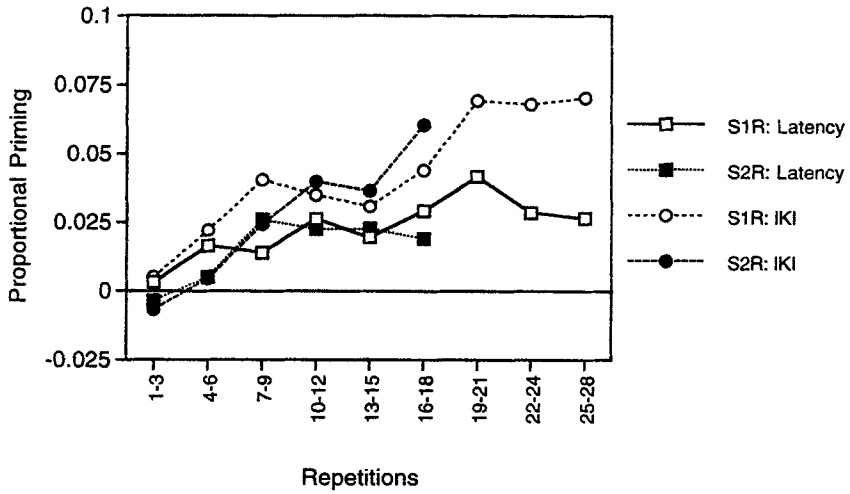


Figure 5. Proportional priming from Experiment 2 across blocks for Session 1 repeat (S1R) and Session 2 repeat (S2R) stimuli. IKI = interkeystroke interval.

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participant were lost due to computer failure, so only 15 participants were available for the analysis). Treating the continuous scale of recognition memory judgments as indicating how confidently the participants judged an item to be "old" as opposed to "new," it was observed that target strings from the original (acquisition) transition rule set were judged as older than distractor strings generated with the same transition rules. Target strings from the complementary transition rule set (new rule) were judged as older than distractor strings generated from the new transition rules. Session 2 and Session 3 uniques, with which participants had experience later in training and hence closer in time to the recognition test than Session 1 uniques, were more significantly discriminable, whereas Session 1 uniques were not. New rule uniques, containing transitions with which participants had little experience compared to the amount of experience they had with transitions from the original (acquisition) rule set, were judged as newer (i.e., less confident that they were seen before) than items generated from the acquisition rule set. Likewise, distractors generated from the new rule set were judged as newer than distractors generated from the original (acquisition) rule set. Thus, recognition performance was driven both by familiarity with whole digit strings and by familiarity with transition rules.

Relationship Between Skill Learning and Priming

The relationship between skill learning and priming across participants was examined as in Experiment 1 by computing skill learning and priming scores, which are presented in Table 3. No correlations between skill learning and priming were significant. This may be due to a greater reliance on whole-string encoding in Experiment 2, which would presumably affect the priming measure to a greater degree than the skill learning measure.

Discussion

In this experiment, as in Experiment 1, both skill learning and priming were observed. The change in procedure for the DET instituted in Experiment 2, which was expected to encourage encoding of whole five-digit strings, succeeded in enhancing priming. As a function of this manipulation of encoding, priming in Experiment 2 was more clearly incremental than in the previous experiment, showing a greater repetition effect with increasing numbers of repetitions, regardless of the point in training at which new repeating strings were introduced. Despite the emphasis on encoding whole five-digit strings, however, performance in Experiment 2 was influenced by knowledge acquired over trials about the digit-to-digit transitions, as it was in Experiment 1. Significant performance decrements were seen here upon introduction of unique digit strings constructed from the complementary rule set (new rule uniques), just as in the previous experiment. And the largest category of errors was again transition-legal errors, involving substitutions of digits consistent with the first-order transition rules. It was also the case, as in Experiment 1, that performance on the new rule uniques was still superior to performance at the start of training on the uniques from the acquisition rule set, indicating that another portion of the learning in the DET was of information common to both rule sets. Thus, the effects of memory were again shown to operate at three distinct levels of organization: single digits, common to both rule sets; digit-to-digit transitions, specific to digit strings generated from one or the complementary rule set; and whole five-digit strings. Skill learning and priming were uncorrelated in Experiment 2, suggesting that memory may have different effects at different levels of organization.

The results from the recognition test indicate that participants have significant explicit remembering about the digit strings encountered during training. Some explicit remember-

ing of whole five-digit strings was demonstrated by participants being more confident in calling repeated strings "old" than strings seen only once, and more confident in calling either category of previously seen strings "old" than strings generated for the recognition test using the same transition rules. There was also evidence that first-order transition information influenced recognition judgments. Participants more confidently recognized uniques from the original (acquisition) rule set than uniques from the complementary rule set, even though each of the uniques was seen only once each in the same session, suggesting that the difference in rule structure between the items affected recognition judgments. In addition, recognition performance differed between distractors generated from the original rule set and distractors generated from the new rule set; participants were more confident that the distractors generated from the original rule set had been seen before than that the distractors generated from the new rule set had been seen before, even though none of these whole items had been presented previously.

The precise relationship between these aspects of explicit remembering and the skill learning and priming effects observed in the DET will need to be explored in further work. However, it should be noted that the effects of first-order transition structure on skilled performance have been observed in amnesic patients, who have impaired explicit remembering of training instances (e.g., Nissen & Bullemer, 1987; Reber & Squire, 1994; but see Curran, 1997), and in college-aged participants who report no awareness for the repeating sequence (Willingham, Nissen, & Bullemer, 1989). These data, along with similar dissociations exhibited by amnesic patients (see above), indicate that explicit remembering is neither a necessary concomitant nor the underlying cause of various aspects of the acquisition of skill.

Experiment 3

The first two experiments demonstrated that priming can occur throughout the early phase of skill learning. Experiment 3 attempted to examine the development of skill during a more extensive range of training, taking novices through 18 sessions of the DET. Although it is unlikely that these novices could be trained up to the level of true "experts" in 18 sessions, this extended training experiment offered further insight into the nature of priming at various levels of skilled performance.

One recurring question regarding the relationship between skill learning and priming is whether response times, or priming scores, or both follow a power function. Logan (1990) demonstrated that response times for repeated items in a lexical decision task decreased as a power function, just as do response times in a large number of skill learning tasks (Newell & Rosenbloom, 1982). Kirsner and Speelman (1996) extended this finding by demonstrating that response times decreased as a power function of both pre-experimental practice (modeled by word frequency) and experimental practice. Repetition priming also decreased as a power function of pre-experimental practice. However, the effects of experimental practice were not captured by a power function; rather, priming was relatively constant across repetitions. As Kirsner and Speelman (1996) noted, these data seemed to contradict Logan's (1988) instance theory of automaticity, which did not originally include a mechanism for improved performance on new items and thus seemed to predict power-function priming scores. However, as we demonstrate below (see *Computational Analysis of Skill Learning and Priming*), a simple modification of Logan's instance theory that allows generalization to new items can produce constant priming across repetitions. To shed further light on the empirical question of whether skill learning and repetition priming are characterized by power function speed-up, we tested the fit of the power function to skill learning and priming data in Experiment 3.

Experiment 3 also investigated whether practice on strings generated from new transition rules would interfere with performance on strings generated from old transition rules, by changing transition rule sets across training and testing on strings generated from each. Work with connectionist network models has shown that networks with fully distributed learning exhibit a form of "catastrophic" interference, in which performance on previously learned items is drastically degraded by training on new items (McCloskey & Cohen, 1989). The human memory phenomenon of explicit remembering does not exhibit this type of forgetting, however. The present experiment explored whether procedural forms of human memory, as exhibited in the DET, may exhibit catastrophic forgetting.

Method

Participants

Four undergraduate students from the University of Illinois at Urbana-Champaign participated in the experiment, which comprised 18 1-hr sessions on separate days. Each of the students volunteered to participate as part of an extended research practicum. (A 5th participant was excluded due to a computer power loss in Session 7, which resulted in the failure to complete the session.) It was emphasized to the participants in this experiment that they were free to withdraw at any time, and they were reminded that they should withdraw from the experiment if they developed any discomfort due to repetitive strain (no participants did so).

Procedure

The procedure for each session, conducted on separate days, closely followed that of the first two sessions of Experiment 2.

Stimuli

The stimuli used during the 18 sessions of Experiment 3 are shown in Table 5. For each set of four sessions, participants were trained on the same set of stimuli. Thus, unique strings were unique within a particular session (i.e., occurred only once in a given day of testing) but appeared once in each session for one or more sets of four sessions. "Acquisition rule unique" (ARU) strings appeared in each of the 18 sessions. Repeat strings were presented repeatedly within a session, appearing once in each of the blocks in a session. "Acquisition rule repeat 1–8" (ARR1-8) strings were presented in

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Table 5
Stimuli Used in Experiment 3

Session(s) and stimulus type	No. per block	Rule set
1–4		
Acquisition rule repeat 1–8	12	1
Acquisition rule unique	36	1
5–8		
Acquisition rule repeat 1–8	12	1
Acquisition rule repeat 5–8	12	1
Acquisition rule unique	24	1
9–12		
Transfer rule repeat 9–16	12	2
Transfer rule unique	24	2
Acquisition rule unique	12	1
13–16		
Transfer rule repeat 9–16	12	2
Transfer rule repeat 13–16	12	2
Transfer rule unique	12	2
Acquisition rule unique	12	1
17		
Acquisition rule repeat 1–8	12	1
Acquisition rule unique	36	1
18		
Transfer rule repeat 9–16	12	2
Transfer rule unique	12	2
Acquisition rule unique	24	1

each block in each of the first 8 sessions, whereas "acquisition rule repeat 5–8" (ARR5–8) strings were presented repeatedly only in Sessions 5–8. In Session 9, unique and repeating strings were introduced that were generated from the complementary (transfer) rule set. "Transfer rule unique" (TRU) and "transfer rule repeat 9–16" (TRR9–16) strings were presented in Sessions 9–16; the uniques occurred only once per session, whereas the repeats occurred in each block in each session. "Transfer rule repeat 13–16" (TRR13–16) strings were presented repeatedly in Sessions 13–16. Unique strings from the acquisition rule set (ARU strings) were also included in Sessions 9–16 (12 per block) to examine whether practice in the new rule condition would interfere with old rule knowledge. In Session 17, participants were presented exclusively with acquisition rule set strings (ARR1–8 and ARU), and in Session 18 they were presented with both new and acquisition rule set strings (TRR9–16 and TRU, and ARU).

Results

Results for the first 16 sessions were analyzed separately in groups of 4 sessions.

Latency

Latency data for all sessions of Experiment 3 are shown in Figure 6.

Skill learning. Skill learning was observed for unique strings across the first eight sessions, linear trend $F(1, 3) = 1{,}323.5$, $MSE = 20{,}015.7$. To further analyze these data, power functions (Newell & Rosenbloom, 1982) were fit to mean latencies using log-log regression. The log-log data, along with the regression line, are shown in Figure 7. The data were fit reasonably well by the power function ($R^2 = .79$), but there was a noticeable lack of fit late in training, where response times continued to decrease to a greater degree than predicted by the power function.

Priming. There was significant priming in each session for latency. These effects were analyzed using contrasts to compare repeated to ARU strings; the results of these contrasts are presented in Table 6. Priming was observed for each

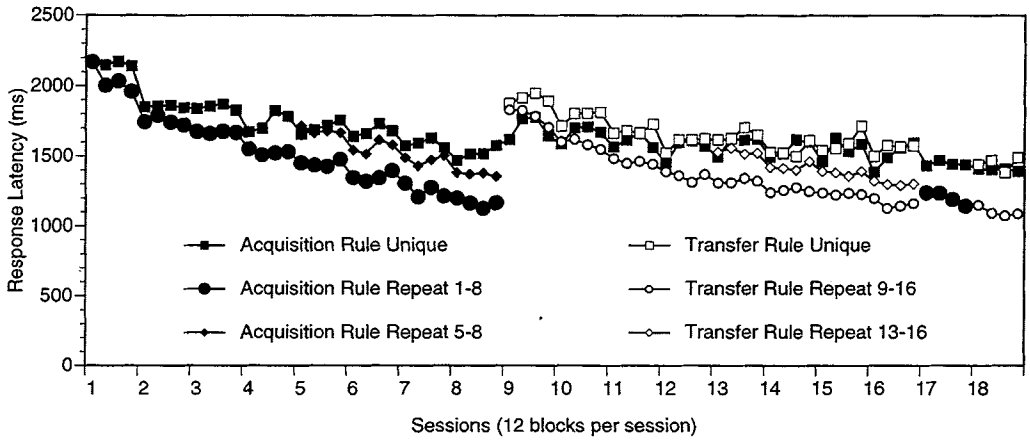


Figure 6. Digit entering latency from Experiment 3 across sessions for each of six stimulus types. Acquisition Rule Repeat 1–8, Acquisition Rule Repeat 5–8, Transfer Rule Repeat 9–16, and Transfer Rule Repeat 13–16 indicate repeating stimuli introduced during Sessions 1, 5, 9, and 13, respectively. Acquisition Rule Unique and Transfer Rule Unique refer to nonrepeated stimuli. Acquisition Rule Repeat 1–8, Acquisition Rule Repeat 5–8, and Acquisition Rule Unique represent stimuli generated from Transition Rule Set 1, while Transfer Rule Repeat 9–16, Transfer Rule Repeat 13–16 and Transfer Rule Unique represent stimuli generated from Transition Rule Set 2. For data presentation only, responses were collapsed over groups of three blocks. See the text and Table 6 for additional details.

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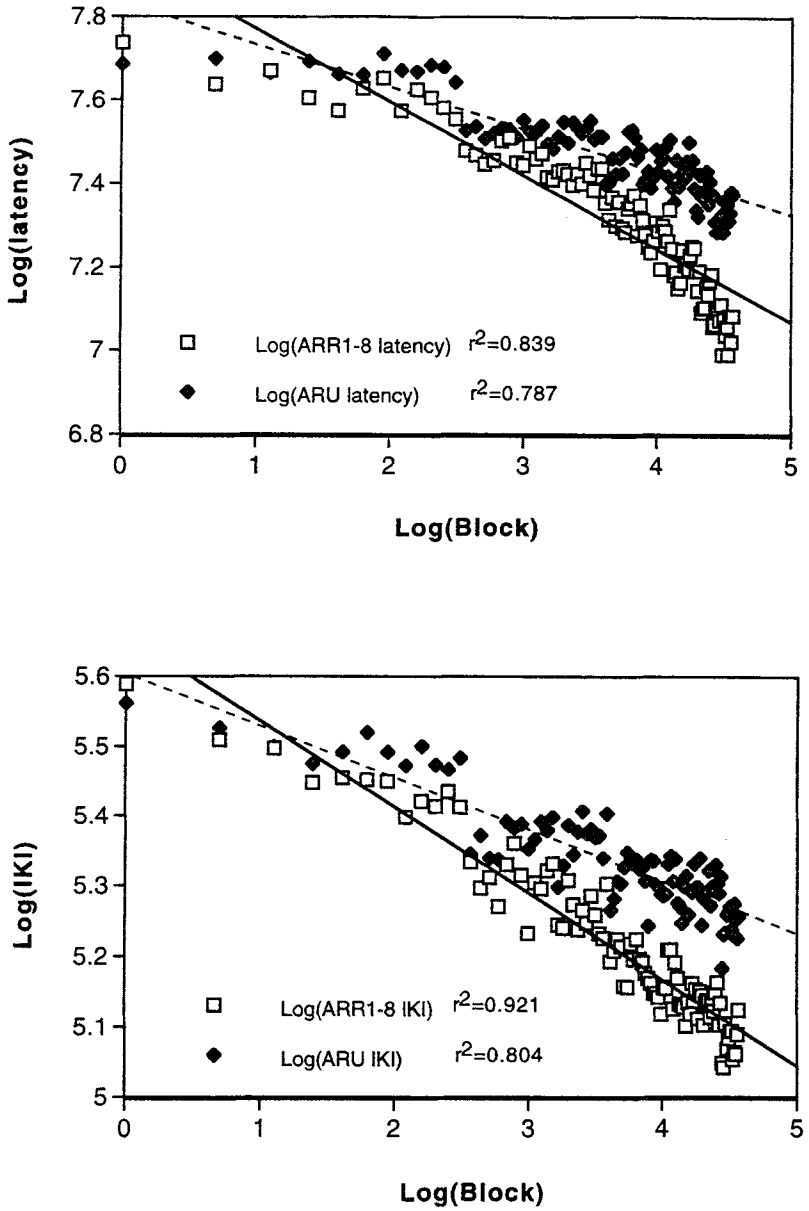


Figure 7. Observed response times (points), fitted power functions (lines), and power function fit correlations for Acquisition Rule Repeat 1–8 (ARR1–8) and Acquisition Rule Unique (ARU) latency and interkeystroke interval (IKI) in Experiment 3.

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set of repeating strings in each set of four sessions. In addition, priming for ARR1–8 strings persisted over eight intervening sessions, as seen by significant priming for ARR1–8 strings in Session 17. Latency for ARR1–8 strings in Session 17 was slightly slower than latency for these strings in Session 8, but this difference was not significant ($F < 1$).

Priming for ARR1–8 strings was further analyzed by fitting a power function to mean latencies for repeating strings and to priming scores (latency for ARU strings over Sessions 1–8 minus latency for ARR1–8 strings). These data, along with regression lines, are presented in Figures 7 and 8 for repeating strings and for priming scores, respectively. Latencies for repeating stimuli were fit well by the power function ($R^2 = .84$), though again latency continued to decrease faster than predicted later in training. Priming scores were not well fit by the power function ($R^2 = .48$).

Transition rule effects. As in the previous experiments, participants were sensitive to digit-to-digit transition information. Introduction of strings generated from the new rule set had a strong detrimental effect on performance for ARU strings; ARU strings had significantly slower latencies in Session 9 than in Session 8, $F(1, 3) = 199.6$, $MSE = 3{,}981$. This effect was overcome by Session 17, at which point latency for these strings was slightly faster than that

Table 6
Results of Planned Comparisons for Priming in Experiment 3

Stimulus type	Latency		IKI	
	$F(1, 3)$	*MSE*	$F(1, 3)$	*MSE*
Session 1 repeat				
Sessions 1–4	32.12*	70,364	19.65*	1,367
Sessions 5–8	11.17*	83,838,251	9.81†	782,657
Session 17	14.04*	102,234	7.89†	1,969
Session 5 repeat				
Sessions 5–8	25.13*	3,673,015	8.67†	126,616
Session 9 repeat				
Sessions 9–12	24.70*	14,138,258	17.13*	140,550
Sessions 13–16	27.73*	39,645,114	11.20*	467,851
Session 18	11.89*	5,778,644	8.98†	64,120
Session 13 repeat				
Sessions 13–16	35.98*	7,983,094	25.75*	72,772

Note. IKI = interkeystroke interval.
*$p < .05$. †$p < .07$.

observed during Session 8 (though only marginally so, $p < .10$).

As tested by contrast across Blocks 9–12, performance on Rule Set 2 (transfer) strings introduced in Session 9 (TRU) was initially slower than on ARU strings, $F(1, 3) = 24.64$, $MSE = 4{,}535{,}537$. However, a significant Trial Type (ARU, TRU) × Block (9–16) interaction in an ANOVA reflected the fact that the difference between latencies for these strings decreased over blocks, $F(95, 285) = 1.57$, $MSE = 12{,}508$. By the end of the experiment this difference was gone; performance for ARU and TRU strings was equivalent across Sessions 13–16 ($F < 1$).

IKI

IKI data for all sessions of Experiment 3 are presented in Figure 9.

Skill learning. Significant skill learning was also observed for IKI for acquisition rule unique strings across the first eight sessions, linear trend $F(1, 3) = 334.9$, $MSE =$

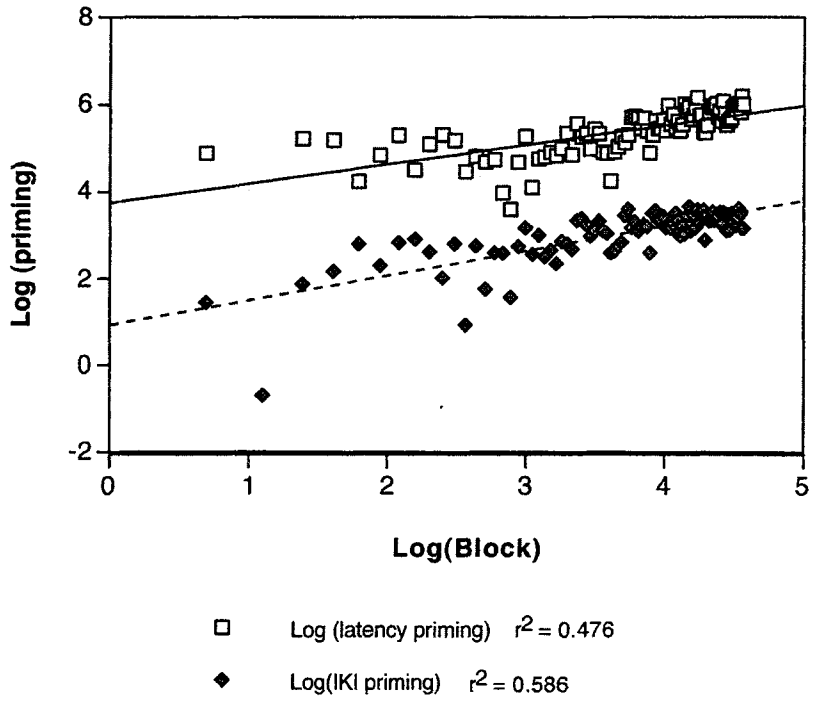


Figure 8. Observed priming (points), fitted power functions (lines), and power function fit correlations for latency and interkeystroke interval (IKI). For each response time measure, priming was computed as (Acquisition Rule Unique − Acquisition Rule Repeat 1–8).

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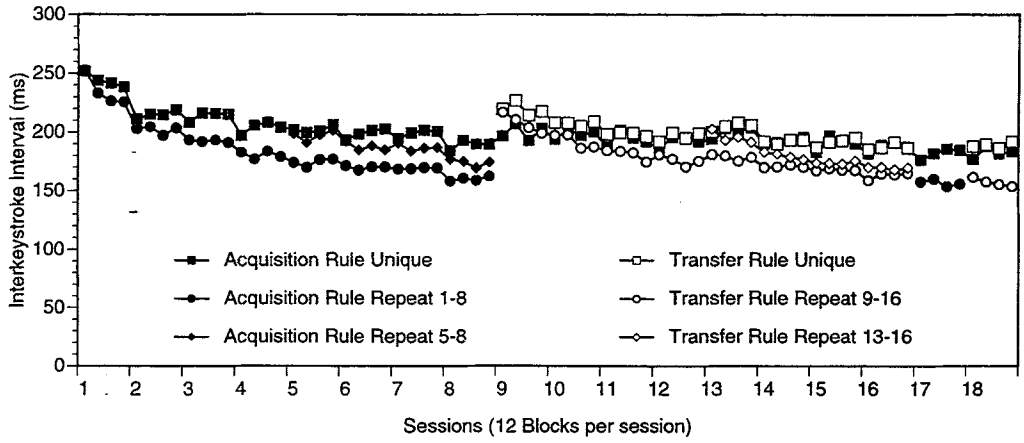


Figure 9. Digit entering interkeystroke interval (IKI) from Experiment 3 across sessions for each of six stimulus types. Acquisition Rule Repeat 1–8, Acquisition Rule Repeat 5–8, Transfer Rule Repeat 9–16, and Transfer Rule Repeat 13–16 indicate repeating stimuli introduced during Sessions 1, 5, 9, and 13, respectively. Acquisition Rule Unique and Transfer Rule Unique refer to nonrepeated stimuli. Acquisition Rule Repeat 1–8, Acquisition Rule Repeat 5–8, and Acquisition Rule Unique represent stimuli generated from Transition Rule Set 1, whereas Transfer Rule Repeat 9–16, Transfer Rule Repeat 13–16 and Transfer Rule Unique represent stimuli generated from Transition Rule Set 2. For data presentation only, responses were collapsed over groups of three blocks. See the text and Table 6 for additional details.

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210.97. A power function was fit to the IKIs for these strings using log-log regression; the data, along with the power fits and power function, are presented in Figure 7. These data were fit relatively well by the power function ($R^2 = .80$).

Priming. Significant priming was observed on IKI for each set of repeating strings in each set of four sessions, as revealed by contrasts. The results of these analyses are presented in Table 6. As with latency, IKI priming for ARR1–8 strings persisted across eight intervening sessions; IKI to ARR1–8 strings in Session 17 was slightly faster than IKI to these items in Session 8, but this difference was not significant ($F < .5$).

Priming for IKI was further analyzed by fitting power functions to mean IKI for repeating strings and to priming scores. Repeating item IKI was fit well by the power function ($R^2 = .92$; see Figure 7). Priming scores were fit less well by the power function ($R^2 = .59$; see Figure 8), apparently due to instability of priming scores early in training.

Transition rule effects. Unlike latency, performance for ARU strings was only marginally affected by the introduction of strings generated from the new rule set (TRU); IKI for these strings was only marginally slower in Session 9 than Session 8, $F(1, 3) = 5.92$, $MSE = 457.2$, $p < .1$.

As tested by contrast across Blocks 9–12, performance on Rule Set 2 (transfer) strings introduced in Session 9 (TRU) was slower than on ARU strings, $F(1, 3) = 84.73$, $MSE = 9{,}129$. As with latency, a significant Item Type × Block interaction was found, $F(95, 285) = 1.44$, $MSE = 116.3$, indicating that the initial difference in performance for unique strings from the two rule sets decreased across blocks. By the end of the experiment, this difference was gone; there was no difference in IKIs for these strings in Sessions 13–16 ($F < 1$).

Accuracy

Analyses of the accuracy data are presented in Appendix A. These analyses largely confirmed the response time analyses, but a significant decrease in accuracy for ARU items was observed across Blocks 1–8, suggesting that a speed–accuracy tradeoff may have occurred for those items.

Discussion

Analyzing data across an extended training regimen, Experiment 3 demonstrated several features of learning and priming in the DET. Skill learning continued throughout training and was fit well by a power function. Priming occurred at all levels of expertise; the increase in speed with repetition was also fit well with the power function, though the increases in priming (measured as a difference score) were not fit well by a power function. These findings of power-function speed-up for unique and repeated items but not for repetition priming replicate the findings of Kirsner and Speelman (1996) and Logan (1990), suggesting that these characteristics hold across a wide range of task domains (from lexical decision to perceptual-motor performance).

It is interesting to note that performance of participants after 18 sessions of training, although much improved over initial performance, was still inferior, by a considerable margin, to that of digit entry "experts"—accountants with an average of 16 years of job-related experience at digit

entry (Poldrack, Selco, Field, & Cohen, 1994). We found that these experts produced latencies in the DET that were almost twice as fast (691 ms) in their first block as were the latencies for unique items for participants in the last block of Session 18 in Experiment 3 (1,394 ms). This suggests that skill acquisition in this task is a very lengthy process, and that the results presented here with an 18-session training regimen still reflect only relatively early phases of skill acquisition in this task.

Experiment 3 demonstrated that repetition effects in this task can remain completely intact over eight intervening sessions of training on different strings. Long-lasting repetition effects in a similar task were reported by Fendrich et al. (1991), who found effects lasting at least 1 month. The present results expand the Fendrich et al. findings by showing that repetition effects last even when the intervening period is filled with training on items with different (largely opposing) sequential structure; the persistence of repetition effects in Fendrich et al. occurred across a period without further training in their task.

One goal of Experiment 3 had been to examine whether transfer to a new set of transition rules would lead to interference with the knowledge of the previously trained transitions. The results are clear in their demonstration that the new rules did interfere with knowledge of the first set of transitions, but this interference was not catastrophic in any sense, and the new knowledge was integrated with the old knowledge fairly quickly. This lack of interference may have arisen from the fact that the new and old rules were presented in an interleaved rather than a massed fashion (Masson, 1990).

Finally, note that although the skill learning data (performance for unique strings) suggested the presence of a speed–accuracy tradeoff, its only impact on the current investigation of the relationship between skill learning and priming was to provide a more conservative test for the presence of priming. That is, if the increase in performance speed (decrease in latency) for unique items here was achieved in part by a corresponding decrease in accuracy, it would only serve to make it harder to observe priming in the latency measures (because the baseline is moving in the same direction as priming). The fact that priming was nonetheless observed across all levels of skill is, therefore, all the more impressive.

General Discussion

The experiments reported here systematically investigated the relationship between skill learning and priming within a single task. Experiments 1 and 2 examined learning over three sessions in the DET in novice participants and found both general skill learning and priming occurring throughout training. In both experiments, the performance gains induced by experience were clearly dependent on knowledge acquired at three different levels of stimulus structure: single digits, digit-to-digit transitions, and whole five-digit strings. The correlation between skill learning and priming varied from significantly negative to nonsignificant, suggesting that the relationship between skill learning and priming (defined as item-general and whole-string learning, respectively) can vary within a single study for different measures.

Experiment 3 extended training in the task to 18 sessions, showing both skill learning and priming, and showing the development of long-term priming that lasted over 8 intervening sessions with interfering stimuli. When new stimuli were introduced that embodied different digit-to-digit transition structure, the new knowledge was seemingly well integrated into the knowledge supporting long-term priming. Response times for unique and repeated items followed a power function, whereas priming scores did not, replicating previous findings.

Item Specificity and the Definition of an "Item"

Most current accounts of memory present priming and skill learning as two forms of implicit, procedural, or nondeclarative memory, but then treat them as separate types of memory, one supporting item-specific effects of experience and the other supporting effects of experience that generalize to other items within the trained domain. However, the results of the present studies are awkward to account for under a strict distinction between skill learning and priming, given the demonstrable one-digit, two-digit, and higher order effects. The current finding of learning at these multiple levels raises the following question: If we reserve the term *priming* for the effects of repetition of whole five-digit strings in this task, and we reserve the term *skill learning* for learning of the one-digit (mapping) information that generalizes to all exemplars in this domain, thereby using these terms as they are used in the literature, then what kind of memory effect applies to the learning of digit-to-digit transition information in the DET?

It is not at all obvious that these various effects differ in their degree of specificity or generality. For example, there is no reason to believe that the learning of the digit-to-digit transitions that repeated throughout training on the acquisition rule set is any less item-specific than the learning of the whole five-digit strings that repeated in the task; yet, such learned information permits performance to generalize to other unique strings containing those same digit-to-digit transitions. Such results would seem to blur the distinction between skill learning and priming, if distinguished solely in terms of the degree of item-specificity of the effects. We suggest, instead, that they may be the same effects of memory measured at different levels of stimulus organization—that is, as the effects of memory operating on items that are defined differently at various levels within the information processing stream. The materials used in the DET were constructed to exhibit certain regularities at the one-digit, two-digit, and whole five-digit levels of organization, and the performance data indicate that memory exerted its effects at each of these levels. There is nothing special about these particular levels of organization. Indeed, there is every reason to believe that if the task had built in regularities at other levels as well, then performance would have shown still further levels at which memory can exert its effects, each one being item-specific at that level. Accord-

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ingly, it seems more reasonable to attempt to discover the various loci of learning effects within the processing stream and then to come to understand the nature of those effects, rather than to attempt to enforce a binary classification into item-specific (priming) and item-general (skill learning) effects.

Skill Learning and Priming Within the Procedural–Declarative Memory Framework

Within the framework of the procedural–declarative distinction, Cohen and Eichenbaum (1993) have proposed a mechanism that integrates the phenomena variously labeled as skill learning and priming. This learning mechanism, called procedural memory, operates in neural processing networks to tune and bias performance so that the network performs most optimally in its immediate environment; that is, it tunes itself to the statistical structure of the environment in response to experience on an item-by-item basis. The knowledge acquired by means of procedural memory is represented through changes in the connection strengths in those neural networks that are involved in processing, and thus can be fruitfully described in terms of connectionist models of learning (e.g., Rumelhart & McClelland, 1986). Procedural memory is contrasted with declarative memory, which relies crucially on the relational memory mechanisms found in the hippocampal system and which supports explicit remembering of new facts and events.

On this view, priming and skill learning are seen as consequences of the same incremental learning mechanism, tied to and expressed through the operation of various neural networks at different levels of organization in the system. Returning to the results of the studies reported here, experience of participants in the DET would produce changes in the network(s) supporting the mapping of the perceptual forms corresponding to digits onto the response operations necessary to enter digits on the keyboard, with these changes capturing all the regularities in the stimulus environment—on the basis of the repetition of the specific one-digit, two-digit, and five-digit regularities that were built into the task.

Computational Analysis of Skill Learning and Priming

Although it is often assumed that the demonstration of dissociation between two classes of phenomena strongly implies that they are supported by separate systems (e.g., Schwartz & Hashtroudi, 1991), this assumption has been called into question by a number of investigators (e.g., Dunn & Kirsner, 1988; Weiskrantz, 1968) on theoretical grounds. To understand the previously discussed dissociations between skill learning and repetition priming, we adopted a computational approach, which attempted to answer the following question: Can these dissociations arise when skill learning and repetition priming arise from the same learning mechanism? We examined this question using two different models: a three-layer connectionist network using the back-propagation learning algorithm, and an extension of Logan's (1988) instance theory of automaticity that allowed for generalization to new items. Although these models differ substantially in the nature of their learning mechanisms, they share the common feature that skill learning and repetition priming arise from a common mechanism within each model. It should be noted that these models were not meant as models of performance on the DET or any specific task (for a model of the DET, see Gupta & Cohen, 1998). They were meant rather as a sort of existence proof, demonstrating that a certain set of dissociations can be accounted for by a single mechanism. Details of the computational models are presented in Appendix B. Our testing of these models was designed to capture two phenomena that have been taken as evidence for a dissociation between skill learning and priming: lack of correlation between skill learning and priming, and differing relationships between skill learning and priming across training (i.e., constant priming with increasing skill learning, or increasing priming with increasing skill learning).

Connectionist Model

Performance of the connectionist model on an autoassociator task (i.e., reproducing an input pattern on the output units) was examined by varying the number of hidden units in the model (8 or 16) and lag between subsequent repetitions of a given item (24, 28, or 96). Skill learning was defined as an increase in performance (i.e., decrease in mean squared error) across training epochs, and repetition priming was defined as the difference in performance between repeated and new items on each epoch. Results of the connectionist model (proportional priming across training epochs) are presented in Figure 10, and results of a correlational analysis between skill learning and priming are presented in Table 7.

Each of the patterns of data previously taken to support independence between skill learning and repetition priming was reproduced by the connectionist model. Figure 10 demonstrates that priming was relatively constant across repetitions with long lag but increased across repetitions with shorter lag. It is of interest to note that Kirsner and Speelman (1996), who found constant priming with repetitions, used a longer lag than other studies that found increasing priming effects (e.g., Logan, 1990). Table 7 demonstrates that the correlation between skill learning and priming in the model varied from nonsignificant (as found for latency and IKI in our Experiment 2 and IKI in our Experiment 1; cf. also Schwartz & Hashtroudi, 1991) to significantly negative (as found for the latency measure in our Experiment 1).

Repetition lag had strong effects on the nature of priming in the connectionist model. One way in which repetition lag might have affected the size of the repetition effect in this model is through the modulation of interference. When lag is short, there is less interference from intervening items than when lag is long. To the extent that interference is reduced, the facilitation for specific items will remain and priming effects will grow. As the lag becomes longer and interference grows, priming effects are reduced because the representa-

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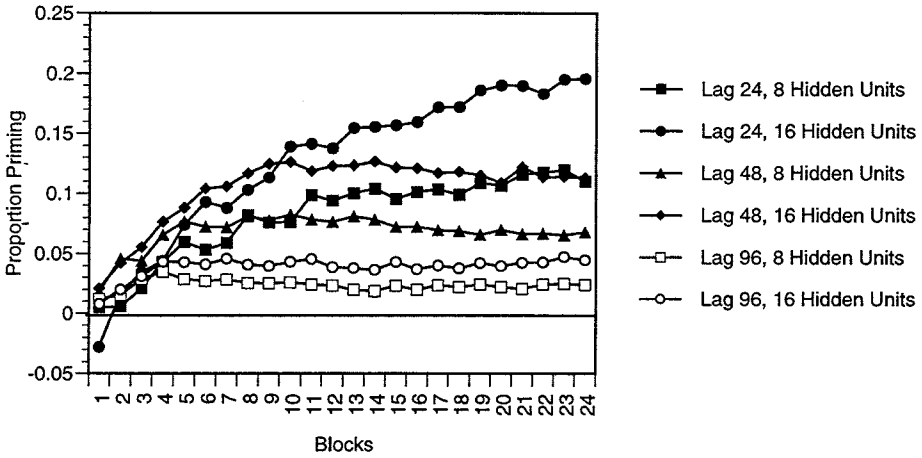


Figure 10. Proportion priming as function of repetitions for the connectionist model.

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tion of repeated items is degraded as new items are stored. The effects of the number of hidden units may also be explained by interference, because a smaller number of hidden units results in a more compact representation; thus, the representations of individual items will exhibit less differentiation and will cause greater interference with each other. This explanation suggests that other manipulations that increase interference (e.g., item similarity) might have similar results on the nature of priming in the network.

Instance Model

Performance of the modified instance model was examined by varying the probability of item generalization (0, 0.5, and 1.0) and the mean of the retrieval time distribution for similar items (525, 550, or 575 ms); see Appendix B for details. Skill learning was defined in terms of the difference between the model's response times across training blocks, and repetition priming was defined as the difference in model response times between repeated and new items. Proportional priming data for the model are presented in Figure 11, and correlational data between skill learning and priming are presented in Table 8. Like the connectionist model, the instance model was able to produce each of the phenomena previously thought to indicate separate skill learning and priming mechanisms. When the probability of generalization to new items was high, priming remained relatively constant across repetitions, whereas priming increased with repetitions when generalization was less likely. In addition, the model produced correlations between skill learning and priming that ranged from nonsignificant (when generalization was unlikely) to significantly negative (when generalization occurred for every new item).

It is interesting to compare the reasons for different relationships between skill learning and priming in the instance theory to those in the connectionist model. Whereas these effects in the connectionist model were thought to arise from interference, in the instance theory model there is no mechanism for interference with stored traces; all traces remain in memory, and each new trace is stored without writing over any existing traces. In the instance theory, different relationships between skill learning and priming arose from differences in the degree that new items were able to retrieve similar items from memory. When these items were able to retrieve many items from memory, priming effects were small and remained constant because the few additional identical traces retrieved by old items only slightly improved performance for those items. When new items were not able to retrieve many traces from memory, priming effects grew quickly because of the statistics of extreme values.

Although this interpretation differs from the interpretation of the connectionist model, the two interpretations converge to suggest that the nature of the stimulus domain can affect the relationship between skill learning and priming. The instance model data suggest that when generalization is common, as when items in the stimulus domain are quite similar, priming should be small and should remain constant with repetitions, whereas when generalization is rare priming should be greater and should increase with repetitions. The effects of varying interference in the connectionist model are directly related to the effects of generalization; in fact, interference can be viewed simply as a case of destructive generalization. Thus, as interference (generaliza-

Table 7
Correlations Between Skill Learning and Repetition Priming for Connectionist Models

Parameter	*r*
8 hidden units	
Lag 24	−.557*
Lag 48	−.355
Lag 96	.074
16 hidden units	
Lag 24	−.612*
Lag 48	−.106
Lag 96	.302

*$p < .05$.

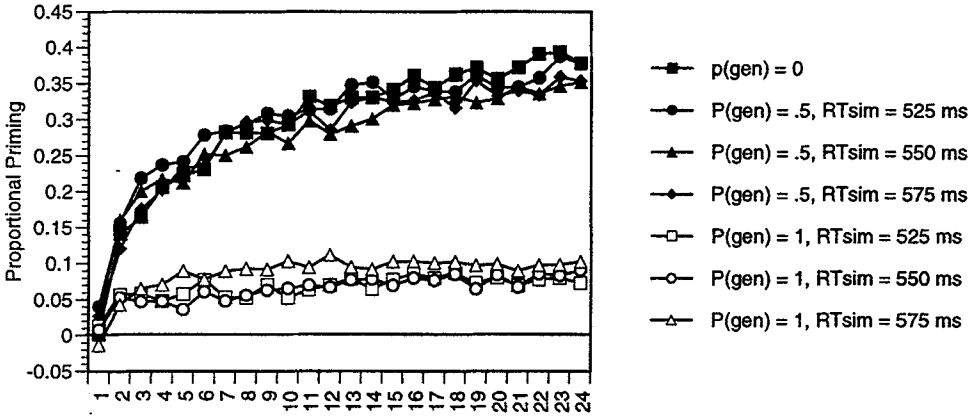


Figure 11. Proportion priming as a function of repetitions for the instance model. P(gen) = probability of generalization; RTsim = mean response time for similar items.

tion) increases, priming effects decrease because new items and old items become more alike. These results suggest that further investigations of the relationship between skill learning and repetition priming could benefit from examination, manipulation of the structure of the stimulus domain, or both.

Reversed Association and Memory Systems

Dunn and Kirsner (1988) have argued that dissociations, even double dissociations, between tasks are not sufficient to demonstrate that those tasks rely on separate memory systems. Instead, they proposed a criterion of *reversed association,* in which a nonmonotonic relationship holds between performance on two tasks when they are measured over at least three conditions. Our single-system models of skill learning and priming provide an independent empirical test of the reversed association proposal; we know that skill learning and priming are produced by a single system, so we should expect a monotonic relationship between skill learning and priming if the reversed association proposal is correct.

The relationship between proportional skill learning and priming scores in the connectionist and instance models are plotted in Figure 12 (A and B) as suggested by Dunn and Kirsner (1988). Several of the specific parameter sets for each of the models exhibited a nonmonotonic relationship between skill learning and priming. In some of these cases, the comparison is compromised by a restricted range of

Table 8
Correlations Between Skill Learning and Repetition Priming for Instance Models

Probability of generalization	Similar item mean *RT*	*r*
0		.029
.5	525	−.406*
.5	550	−.278
.5	575	−.168
1.0	525	−.533*
1.0	550	−.554*
1.0	575	−.500*

Note. RT = reaction time.
*$p < .05$.

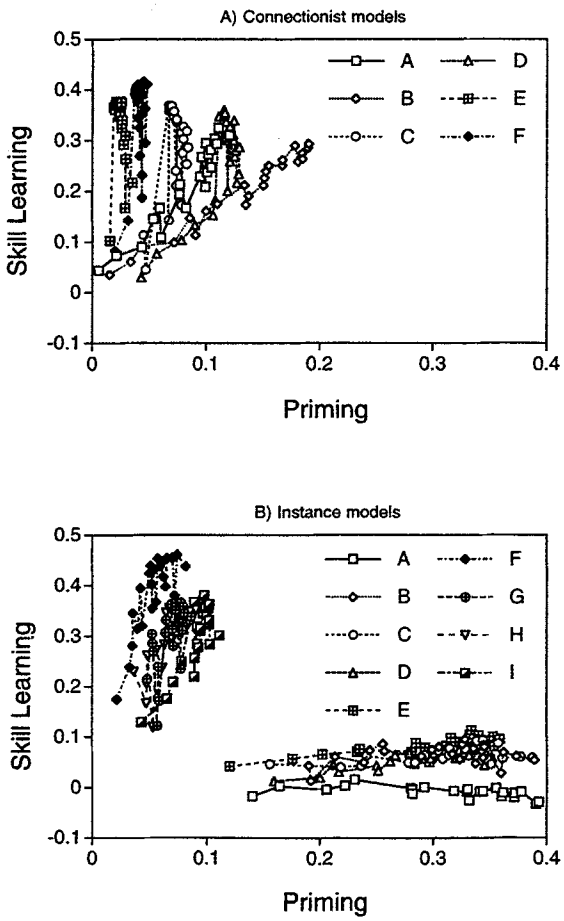


Figure 12. Priming plotted against skill learning as proposed by the reversed association analysis of Dunn and Kirsner (1988) for (A) connectionist models and (B) instance models. A–I are models.

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priming or skill learning scores, but in at least some of the models the nonmonotonicity extends over a range of both skill learning and priming scores. These data point to one difficulty in implementing the reversed association analysis: No statistical test of monotonicity was provided by Dunn and Kirsner. It is thus difficult to determine whether the degree of nonmonotonicity observed in the data is due to sampling error or true nonmonotonicity in the underlying distribution. Our findings suggest that reversed association may arise from a single system, but further work is required to determine whether single systems can produce data that clearly indicate reversed association.

Further Directions for Studying Skill Learning and Priming

The results presented here demonstrate that skill learning and priming may exhibit a relationship that varies across different sets of task demands. This suggests that it may not be fruitful to speak of *the* relationship between skill learning and priming, because this relationship may vary. Such a perspective has implications for the study of these phenomena. The primary implication of the present work is that skill learning and priming can, and must, be studied in the same participants using the same task. Such an approach could clarify the relationship between skill learning and priming in various pathological conditions, such as Huntington's and Parkinson's diseases, where studies heretofore have examined skill learning and priming in different tasks.

The present experimental and modeling results further clarify the kinds of relationships between skill learning and priming that might arise from a single mechanism. On the basis of these results, it no longer seems reasonable to claim that behavioral dissociations between skill learning and priming necessitate the postulation of multiple systems underlying performance. Such a claim requires further evidence to be supported. For example, convergent dissociations in normal people and people with neurological disorders, demonstrating that skill learning and priming can be dissociated within the same task both in normal people and between patient groups, would be necessary before we would want to conclude that skill learning and priming are supported by separate cognitive and neural systems.

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(Appendixes follow)

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Appendix A

Accuracy and Error Analysis

Accuracy was measured with respect to correctly entering all five digits of a given string. The nature of the errors was analyzed according to a taxonomy of error types in line with those used in the transcription typing literature (e.g., Grudin, 1983). *Transition-legal errors* were defined as the entering of digits that deviated from the digits present in the stimulus string but were consistent with the first-order transition rules for strings seen by that participant. *Keyboard errors* were defined as the entering of digits corresponding to keyboard positions that were horizontally or vertically adjacent to the correct ones (e.g., errors made by pressing Keyboard Positions 2 or 4 in response to the digit 1).[A1] In addition, because some errors could fit into both the transition-legal error and keyboard error categories simultaneously, we also identified *nonkeyboard transition-legal errors,* referring to the entry of digits consistent with the transition rules and corresponding to keyboard positions not adjacent to those of the target digits. Finally, *transposition errors* were defined as the entering of a sequence of two digits anywhere in a string that was a transposition of the sequence present in the string (e.g., entering 19247 for 19427); transposition errors could also fit into the aforementioned categories, depending on the substituted transition. Errors were analyzed for all participants in Experiments 1 and 2. The overall number of errors for Experiment 3 was too low to permit any meaningful analyses of error types in that experiment.

Experiment 1

Accuracy data for Experiment 1 are presented in Table A1. A Block × Trial Type ANOVA was performed on accuracy values separately for each session (12 × 2 in Session 1, 12 × 3 in Session 2, and 12 × 4 in Session 3). In Session 1, there was no difference between accuracy for S1R strings and unique strings ($p > .45$). In Session 2, there was a main effect of trial type, $F(2, 30) = 6.58$, $MSE = 0.003$; planned comparisons showed that S1R strings were entered more accurately than both S2R strings and unique strings, which did not differ from each other ($p > .25$). In Session 3, there was also a significant effect of trial type, $F(3, 45) = 19.59$, $MSE = 0.015$. Planned comparisons revealed that S1R and S2R strings were entered more accurately than unique strings, which in turn were entered more accurately than new rule unique strings. A test for linear trend in accuracy for S1R stimuli over the 36 blocks was nonsignificant ($p > .35$); however, there was a significant linear decrease in unique accuracy, $F(1, 15) = 17.49$, $MSE = 0.005$.

Table A1
Accuracy for Each Condition Across Sessions in Experiments 1 and 2

Item type	Session 1	Session 2	Session 3
Experiment 1			
Acquisition rule unique	.89	.88	.87
Session 1 repeat	.90	.90	.91
Session 2 repeat		.89	.91
New rule unique			.83
Experiment 2			
Acquisition rule unique	.90	.91	.91
Session 1 repeat	.92	.93	.92
Session 2 repeat		.91	.94
New rule unique			.88

Analyses on error data were conducted on an average (±range) of 259 (±33) errors committed over 159 (±19) error trials per participant ($N = 16$). Table A2 contains the percentages of the various error types as a function of session. To assess whether there were significant differences in the percentage of error types as a function of session, we performed a 3 (session) × 4 (error type: transition-legal, keyboard, nonkeyboard transition-legal, transposition) ANOVA on the error type proportions. This analysis revealed a main effect of error type, $F(3, 45) = 119.03$, $MSE = 12{,}149.7$. The main effect of session was nonsignificant ($F < 1$), as was the Error Type × Session interaction, $F(6, 90) = 1.63$, $MSE = 95.7$, $p > .15$. Contrasts revealed that transition-legal errors occurred more often than all other error types, $F(1, 15) = 406.84$, $MSE = 236.1$; there was a trend toward greater proportions of keyboard errors relative to nonkeyboard transition-legal errors, $F(1, 15) = 4.33$, $MSE = 74.8$, $p = .055$, and there were fewest transposition errors, $F(1, 14) = 238.51$, $MSE = 412.3$.

Experiment 2

Accuracy data for Experiment 2 are presented in Table A1. A Block × Trial Type ANOVA was performed on accuracy values separately for each session (12 × 2 in Session 1, 12 × 3 in Session 2, and 4 × 4 in Session 3). In Session 1, there was no difference between repeated and unique strings ($p > .2$). In session two, the main effect of trial type was significant, $F(2, 30) = 4.85$, $MSE = 0.006$. Planned comparisons showed that S1R accuracy was greater than S2R accuracy, $F(1, 15) = 5.13$. S1R items were entered more accurately than unique strings, $F(1, 15) = 8.90$, whereas S2R strings were not ($p > .47$). In Session 3, there was a main effect of trial type, $F(3, 45) = 3.01$, $MSE = 0.010$. However, planned comparisons comparing S1 and S2 items to unique items, and comparing unique to new rule unique items, were not significant ($ps > .1$). Tests for linear trend performed separately for unique and S1R stimuli were both nonsignificant ($ps > .2$).

Analyses on error data were carried out on an average of 162 (±20) errors committed over 102 (±14) error trials per participant ($N = 16$). Error data are presented in Table A2. To assess whether there were significant differences in the proportion of error types as a function of session, we performed a 3 (session) × 4 (error type: transition-legal, keyboard, nonkeyboard transition-legal, transposition) ANOVA on the error type proportions. This analysis revealed a main effect of error type, $F(3, 45) = 47.68$, $MSE = 11{,}893.5$. The effect of session was nonsignificant, $F(2, 30) = 2.18$, $MSE = 268.2$, $p = .13$, as was the Error Type × Session interaction, $F(6, 90) = 1.02$, $MSE = 92.1$, $p = 0.42$. Contrasts revealed that transition-legal errors occurred more often than all other error types, $F(1, 15) = 569.10$, $MSE = 170.6$; there were equal numbers of keyboard and nonkeyboard transition-legal errors, $F(1, 15) = 1.48$, $MSE = 57.1$, $p = .24$; and there were fewest transposition errors, $F(1, 15) = 185.3$, $MSE = 460.7$. Thus, despite the

[A1] In analyzing the Lessenberry (1928) transcription typing error corpus, Dvorak, Merrick, Dealey, and Ford (1936; cited in Grudin, 1983) found that the most common error type was a character substitution from a letter in an immediately adjacent row (43%) or column (15%). Errors to diagonally positioned keys occurred only rarely, a finding corroborated in the error analyses in the present study, and hence were not included in the present computation of keyboard errors.

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Table A2
Mean Percentage of Error Types (and Standard Errors) for Experiments 1 and 2

Error type	Session 1		Session 2		Session 3	
	M	*SE*	*M*	*SE*	*M*	*SE*
	Experiment 1[a]					
Transition-legal	50.2	2.4	52.7	1.9	51.7	1.9
Keyboard	37.4	2.5	32.1	2.0	36.0	1.3
Nonkeyboard transition-legal	26.9	2.6	32.1	2.3	28.6	1.8
Transposition	14.6	2.0	12.4	2.1	11.9	1.7
	Experiment 2[b]					
Transition-legal	49.9	1.9	54.7	1.7	57.9	4.5
Keyboard	34.7	1.8	32.5	2.3	37.3	2.8
Nonkeyboard transition-legal	30.3	2.5	32.2	2.3	36.3	3.5
Transposition	17.4	2.5	14.2	1.6	15.6	2.8

[a]Based on 576 trials per session. [b]Based on 576 trials for Sessions 1 and 2, and on 192 trials for Session 3.

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differences in methods between Experiments 1 and 2, participants performing the DET exhibited the same distribution of error types.

Experiment 3

Accuracy data for Experiment 3 are presented in Table A3. Unless mentioned otherwise, these data were analyzed with a repeated measures ANOVA over each group of four sessions. The first set of analyses examined whether accuracy differed for repeat as compared to unique strings in each set of four sessions. Accuracy for repeat strings was greater than accuracy for unique strings in Sessions 1–4 (ARR1–8), $F(1, 3) = 11.67$, $MSE = 0.0004$; Sessions 5–8 (ARR1–8), $F(1, 3) = 16.39$, $MSE = 0.0015$, and (ARR5–8), $F(1, 3) = 7.61$, $MSE = 0.0015$; Sessions 9–12 (TRR9–16), $F(1, 3) = 26.58$, $MSE = 0.0003$; and Sessions 13–16 (TRR9–16), $F(1, 3) = 8.34$, $MSE = 0.0027$.

To determine whether priming was still evident for repeated strings from the acquisition rule set after 8 sessions of exposure to strings from the transfer rule set, we performed an analysis using Session 17 accuracy data. A marginally significant difference in accuracy for ARR1–8 versus ARU strings was obtained in Session 17, $F(1, 3) = 7.34$, $MSE = 0.0007$, $p < .08$; this effect accords with the response time data in demonstrating long-term repetition effects for ARR1–8 strings. In Session 18, accuracy for TRR9–16 strings exceeded accuracy for TRU strings, $F(1, 3) = 13.03$, $MSE = 0.001$. Accuracy for acquisition and transfer rule set unique strings (ARU and TRU) did not differ ($p > .5$).

Transition Rule Effects

To examine the effect of introducing strings generated from the transfer rule set, we carried out additional planned comparisons on the accuracy data from Sessions 9–16. ARU string accuracy exceeded that observed for TRU strings during Sessions 9–12, $F(1, 3) = 12.69$, $MSE = 0.0003$, indicating that participants had acquired knowledge of the acquisition rule set. However, this difference in accuracy between TRU and ARU strings disappeared by Sessions 13–16 ($p > .9$).

Changes in Accuracy Over Sessions

To determine whether accuracy changed over acquisition sessions, we performed analyses for ARU, TRU, ARR1–8, and TRR9–16 strings, specifically addressing whether a linear trend was present. Accuracy for ARR1–8 stimuli remained unchanged ($p > .9$), whereas accuracy for TRR9–16 strings increased across sessions, $F(1, 3) = 7.12$, $MSE = 0.0006$. Performance for ARU stimuli decreased over sessions, $F(1, 3) = 11.04$, $MSE = 0.0010$, whereas performance for TRU strings did not change over sessions ($p > .9$).

Table A3
Accuracy for Experiment 3

Item type	Session																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
ARU	.93	.93	.92	.93	.88	.90	.88	.88	.93	.94	.91	.93	.91	.90	.90	.89	.89	.89
TRU	—	—	—	—	—	—	—	—	.89	.91	.90	.92	.90	.91	.89	.90	—	.87
ARR1–8	.93	.95	.95	.96	.93	.96	.93	.95	—	—	—	—	—	—	—	—	.94	—
ARR5–8	—	—	—	—	.89	.94	.91	.95	—	—	—	—	—	—	—	—	—	—
TRR9–16	—	—	—	—	—	—	—	—	.91	.94	.94	.95	.94	.95	.95	.96	—	.95
TRR13–16	—	—	—	—	—	—	—	—	—	—	—	—	.90	.92	.91	.93	—	—

Note. Dashes indicate that data are not applicable. ARU = acquisition rule unique; TRU = transfer rule unique; ARR1–8 = acquisition rule repeat in Sessions 1–8; ARR5–8 = acquisition rule repeat in Sessions 5–8; TRR9–16 = transfer rule repeat in Sessions 9–16; TRR13–16 = transfer rule repeat in Sessions 13–16.

(Appendix B follows on next page)

Appendix B

Details of Computational Models

Connectionist Model

A three-layer feedforward connectionist network was used in the simulations reported here (Rumelhart & McClelland, 1986). The program used to implement the model was a modified version of the McClelland and Rumelhart (1989) software. The model had 48 input units and 48 output units, and the number of hidden units was varied across simulations between 6 and 18. All units in the input layer were connected to all hidden units, and all hidden units were connected to all output units. Each epoch of training involved the presentation of all training patterns (in random order), each of which consisted of vectors of binary values randomly taking the values of one or zero. When each pattern was presented, the activation from the input units was propagated to hidden units and then to output units. The activation of each unit was a nonlinear (logistic) function of the net input to that unit:

$$\text{activation}_i = \frac{1}{1 + e^{-(\text{net input}_i)}},$$

where net input is defined as

$$\text{net input}_i = \sum_j \text{activation}_j \cdot \text{weight}_{ij}.$$

Weight changes were computed using the backpropagation method (Rumelhart, Hinton, & Williams, 1986) after the presentation of every pattern (updating after each epoch did not qualitatively alter the pattern of results). At the beginning of training, weights were initialized with random weights (between −2 and 2); this was the primary source of between-subjects variation in the model.

For each set of parameters, 24 pseudoparticipants were tested with the given stimulus set. To simulate learning in the task, we trained and tested the model for 24 epochs for each pseudoparticipant. In each block, the set of training items was presented once; half of the items presented in each block were new and the other half repeated throughout training. Items were presented in random order in each block. Six simulations were performed, manipulating number of hidden units (8 or 16) and average repetition lag (24, 48, or 96) factorially. Performance of the network was measured in terms of the mean squared error between desired output and output layer activation; use of a different performance measure (entropy) yielded similar results, suggesting that the results are not an artifact of the particular performance measure.

Instance Model

In the instance theory of automaticity (Logan, 1988), performance is determined by a race between memory retrieval and completion of an algorithm. The original formulation of the theory assumed that the time taken by the algorithm remained constant with practice, primarily for mathematical convenience. However, this limited the ability of the theory to account for task-general learning, such as the learning seen for new items in the previous simulations. This issue has not been directly addressed in the development of the instance theory, so it was necessary to extend the theory to account for learning on new items (although since the completion of the present work, an extension of the instance model to account for generalization has been presented by Palmeri, 1997). To account for learning on new items in a task, the theory has to allow either generalization from old instances to new stimuli or general speed-up in the algorithm for task performance. Because the latter essentially changes the theory to a two-factor theory of skill learning and repetition priming, the former was chosen as a way to examine the relationship between priming and skill learning in the task. On a proportion of trials with new items, old and new items were allowed to retrieve similar instances from memory; performance on these items was then determined by a race between memory retrieval and the (constant) algorithm finishing time.

The degree to which new items can benefit from old instances should depend on the similarity of the new items to the older items. The approach to similarity in the present model was to assume that trace similarity is uniformly distributed (i.e., a rectangular density function). This is not an assumption about the distribution of the similarity space, but rather a binary recoding of the similarity space into similar–nonsimilar; the distribution specifies how many of the other items in memory are similar to the current item, where the metric of similarity remains undefined. The choice of the rectangular distribution was made for practical rather than theoretical reasons. On trials in which a new item was allowed to retrieve instances from memory (the proportion of which is a parameter in the model), the number of traces retrieved was determined by a rectangular generalization distribution with mean $N/2$, where N is the total number of traces in memory. Thus, each item that was allowed to retrieve similar items from memory could retrieve any number of similar items from zero to the total number in memory. It was also assumed that the retrieval of similar items would take slightly longer than would the retrieval of identical items.

The instance model took as parameters the mean and standard deviation of retrieval time distributions for identical trace retrieval, similar trace retrieval, and algorithm finishing time, and provided predicted RTs. In addition, the revised version of the instance theory with generalization took as a parameter the probability that a new item would retrieve instances from memory. The simulations reported here were performed as follows.

1. Specify means and standard deviations for each normal (Gaussian) distribution in the model: (a) identical trace retrieval—time for retrieval of traces for previously encountered exemplars of the item on the present trial, (b) similar trace retrieval—time for retrieval of traces for previously encountered items that are not the same as the item on the present trial, and (c) algorithm finishing time—time for the algorithm to finish. For each pseudoparticipant, the means and standard deviations for each of these normal distributions were determined by sampling from a rectangular distribution (i.e., uniformly distributed variation); these rectangular distributions are referred to as the *sampling distributions.* The sampling distributions used to determine the identical trace retrieval time and algorithm finishing time distributions remained identical across pseudoparticipants; for the mean of the normal distributions, the sampling distribution ranged from 400 ms to 600 ms (with a mean of 500 ms), and for the standard deviation of the normal distributions, the sampling distribution ranged from 50 ms to 150 ms (with a mean of 100 ms). The sampling distribution for mean of the similar trace retrieval distribution was varied across simulations, with means of 500, 525, 550, and 575 ms. In each case, the sampling distribution for the mean ranged from 200 ms wide, and the sampling distribution for the standard deviation ranged from 50 to 150 ms with a mean of 100 ms.

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2. The probability of similar item retrieval was varied across simulations, and was set to 0, 0.5, or 1.0. This parameter determined the probability that, on any given trial, similar instances would be retrieved from memory, and it applied equally to new and repeated items. When this parameter was equal to zero, the model was identical to the instance theory model presented by Logan (1988).

3. Each pseudoparticipant received 24 blocks of training, and 24 pseudoparticipants were tested in each simulation. On each block of training, the model was presented with 24 new items and 24 repeated items (in random order).

4. On each trial, predicted response time was determined by a race between algorithm finishing time and minimum memory retrieval time. For each item, a sample was taken from the algorithm finishing time distribution. For each old item, N_i samples were taken from the identical retrieval time distribution, where N_i is the number of identical traces for that item in memory. This is specified as

$$N\text{old}_i^{\text{Identical}} = N_i.$$

In addition, if that old item was allowed to retrieve similar items from memory (with a probability equal to the generalization parameter), then the number of similar items retrieved was determined by a rectangular distribution ranging from 0 to $N_{\text{total}} - N_i$), where N_{total} is the total number of traces in memory. This is specified as

$$N\text{old}_i^{\text{Similar}} = R(0,1) \cdot (N_{\text{total}} - N_i),$$

where $R(0,1)$ is a uniformly distributed variate ranging from 0 to 1.0. The subtraction of N_i assures that a trace in memory is not retrieved as both an identical and similar trace. For each new item, the number of similar items retrieved was determined by a rectangular distribution ranging from 0 to N_{total}. This is specified as

$$N\text{new}_i = R(0,1) \cdot N_{\text{total}}.$$

Algorithm finishing time, identical trace retrieval time, and similar trace retrieval time were sampled from the normal distributions specified above. Predicted response times for old and new items were derived from memory retrieval times and algorithm finishing times as follows:

$$RT\text{new}_i = \min(similar_{N\text{new}_i}, algorithm)$$

$$RT\text{old}_i = \min(similar, identical, aglorithm)$$

where *similar* and *identical* are samples from the distributions for similar and identical trace retrieval times, respectively.

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